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Contextual Areas

Work More Tomorrow: Resolving Present Bias in Project Management

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Abstract. Project management is responsible for almost 30% of the world's economic activity, with an annual value of \$27 trillion. Traditionally, the frequent late delivery of projects is attributed to Parkinson's Law, which incorporates laziness, procrastination, and self-protection against reduced deadlines in the future. Incentive schemes are widely designed and implemented to eliminate Parkinson's Law. Yet many projects are nonetheless delivered late. To explain this, we show computationally that a significant part of project delay is not explained by Parkinson's Law but can be explained by time-inconsistent behavior, that is, present bias. We therefore propose a new incentive scheme that directly addresses present bias in projects. Our work is among the first to develop a multiperiod model of time-inconsistent preferences for projects. Using concepts from savings theory, we design a Work More Tomorrow (WMT) incentive scheme to mitigate the effect of present bias in project execution. The WMT scheme extends the underlying economic theory of present bias to consider network effects that arise in projects. We show, both theoretically and computationally, that the WMT scheme outperforms the most widely used benchmark incentive scheme by delivering significant improvements in on-time frequency and expected project tardiness at lower cost. Moreover, these improvements increase with project size. An extensive sensitivity analysis reveals the characteristics of projects that respond best to the WMT scheme. We also identify several managerial insights that arise from viewing the frequent delays in project management practice through the lens of a multiperiod model of time-inconsistent behavior.

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Keywords: project management • late delivery • present bias • incentive design

1. Introduction

A project is a “temporary endeavor undertaken to create a unique product or service” (Project Management Institute 2013, p. 3). The global impact of project management within the world's economic activity is estimated at 30% (Hu et al. 2015) with an annual value of about \$27 trillion. Since 1996, membership in the Project Management Institute has increased from about 50,000 to more than 500,000. Project management applications now range from construction and engineering into information technology, research and development, new product and service development, and corporate change

management. A research agenda for project management appears in Hall (2016).

Our work focuses on the pervasive and financially damaging problem of late delivery of projects. There are several factors that make the on-time delivery of projects particularly challenging. One factor is the uncertainty surrounding the duration of the component tasks in a project, especially because of the uniqueness of projects. Another factor is the autonomy that is typically given to project teams in order to encourage creativity and improve the quality of the work experience. As a result of these factors, “project performance is particularly

susceptible to the *procrastinating behavior* of individual workers” (Wu et al. 2014, p. 1299). This behavior has typically been characterized as due to Parkinson’s Law (Parkinson 1955, 1958), whereby work expands to fill the time until its deadline.

Parkinson’s Law is extensively discussed in project management texts (Wysocki 2009, Klastorin 2012). Psychological motivations that support Parkinson’s Law are documented by Krakowski (1974). These include appearing busy to impress management and avoidance of having resources withdrawn or less time allocated in the future as a result of outperforming estimates. Other factors include a tendency toward procrastination and obsession over small details. Gutierrez and Kouvelis (1991, p. 991) model Parkinson’s Law using a work expansion factor that represents “leisure time or unnecessary expansion introduced.” They develop a model of stochastic task completion times and use it to develop managerial insights about the setting of deadlines for tasks. Williams (1995) observes that, the faster progress is made on a task, the greater is the extent of later slowing down because of Parkinson’s Law.

The problem of Parkinson’s Law has been addressed, although not fully resolved, through performance incentive schemes. Rose and Manley (2011, p. 765) provide an overview of incentive schemes that are used in construction projects. They comment that, “not only are financial incentives necessary to enhance motivation at personal and organizational levels, but also to promote unified motivation across . . . project teams.” Both the academic (Howard et al. 1997, Bower et al. 2002, Chen and Hall 2021) and business (Corcione 2019) literatures consistently argue that performance incentives improve project outcomes.

Although incentive schemes are widely designed and implemented to mitigate Parkinson’s Law, many projects are delivered late. For example, 60% of 25,000 software projects during 2011–2015 are reported as late (Standish Group 2015). We therefore suggest, and demonstrate computationally, an alternative explanation for late project delivery, which is project workers’ *time-inconsistent preferences*, that is, *present bias*, as discussed in behavioral economics. We review this literature. Akerlof (1991, p. 1) describes procrastination as occurring when “present costs are unduly salient in comparison with future costs.” This results in a delay that may repeat at subsequent decision points. He provides examples in various decision-making situations, including major purchases, savings, irrational obedience, gang behavior, and politics. Within the context of project management, Wu et al. (2014) use the present bias concept to develop a two-period model of a project without task precedence structure but with either a single worker or team of workers. The result of salience is procrastination in the first period and back-loaded effort in the second period. A possible solution is to

reward work in the earlier stages of a project. Our work, however, models time-inconsistent preferences over multiple periods and multiple precedence-constrained tasks within a general project network.

Phelps and Pollak (1968) propose a model of intertemporal preferences, which Laibson (1994) extends to model time-inconsistent taste for immediate gratification. This model characterizes present bias using two parameters (δ, β), where δ describes time-consistent discounting and β describes *bias for the present*. O’Donoghue and Rabin (1999a) distinguish decision makers who are aware of their present bias from others who are not and show how the level of awareness results in different decisions across multiple periods.

O’Donoghue and Rabin (1999b) study the design of incentives for time-inconsistent procrastinating agents under asymmetric information. Delay is costly to the principal; however, the agent faces costs associated with performing the task efficiently. They show that, for time-consistent agents, the optimal incentive scheme exactly reflects the principal’s true cost of delay. Whereas, for procrastinators, the principal should punish delay more severely than its true cost in order to offset the tendency toward procrastination. Even when the true cost of delay is constant over time, optimal incentive schemes for procrastinators typically involve increasing punishment for delay over time. Our work differs in several ways. First, we assume full information, that is, the project manager regularly observes the progress of the task operator. Second, in project management, the true cost of delay is often not a constant or even a smooth function. Third, in project management, performance is measured by the interactive performance of many tasks within a project network, not performance on a single task. Fourth, the stochastic task cost in their model is observable before decisions are made, whereas the stochastic project progress in our model is only observable after the commitment of effort. Finally, following typical project management practice known as crashing, we allow performance expectations for later tasks to be adjusted in response to earlier delays.

Thaler and Benartzi (2004) propose a plan, *Save More Tomorrow*, that encourages consumer savings by mitigating the present bias issue that commonly reduces savings rates. Our work draws on an analogy between the cost of savings experienced by an employee and the effort required by a project worker to complete a task. Despite the fact that the network structure of a project creates additional complications in evaluating the effect of time-inconsistent behavior at the task level, we incorporate concepts of the savings model into our study of present bias and an incentive scheme to mitigate it. Recent academic work on time-inconsistent behavior includes Li et al. (2017) on optimal stopping under present bias preferences. Also, Liao and Chen (2021) propose a conditional cash transfer program to mitigate the effect of time-inconsistent preferences.

We summarize the contributions of our work. Various incentive schemes have been used in projects to reduce Parkinson's Law, but many projects are still delivered late. Therefore, based on a computational study, we identify and evaluate an alternative cause of delay, which is time-inconsistent preferences, or present bias. Our work is among the first to develop a multiperiod model of time-inconsistent preferences for projects described by general acyclic networks. Using concepts from the economic theory of consumer savings, we design a Work More Tomorrow (WMT) incentive scheme to mitigate present bias. The WMT scheme extends the underlying economic theory to consider network effects that arise in projects from the performance of multiple interacting tasks. We show, both theoretically and computationally, that the WMT scheme outperforms the most widely used benchmark incentive scheme by delivering significant improvements in on-time frequency and expected tardiness at lower cost. Moreover, these improvements increase with project size. An extensive sensitivity analysis reveals the characteristics of projects that respond best to the WMT scheme. We also identify several managerial insights that arise from viewing the delays that are typically observed in project management execution through the lens of time-inconsistent behavior.

This paper is organized as follows. Section 2 introduces our notation, formally defines the problem studied, and specifies our assumptions. Section 3 estimates how much of the late delivery of projects can be attributed to the project worker's present bias. Section 4 describes the design of the WMT incentive scheme, and describes a benchmark scheme used in practice. Sections 5 and 6 provide theoretical results about the relative performance of our incentive scheme and this benchmark, and computational studies that support those results. In Section 5 we consider a single task, whereas in Section 6 we consider a serial network. Section 7 provides a similar computational comparison for a general acyclic project network. We also provide a sensitivity analysis for the improvement offered by our incentive scheme. Section 8 contains a conclusion, several managerial insights, and suggestions for future research.

2. Preliminaries

In this section, we provide background about present bias, define the environment studied, specify our assumptions, and model the connection between operator effort and task performance. Present bias describes a behavioral tendency to overvalue *first period* payoffs and undervalue future payoffs. The most widely used mathematical model of present bias is the quasi-hyperbolic utility discounting model of Phelps and Pollak (1968). This model is

$$U^t(u_t, u_{t+1}, \dots, u_T) = u_t + \beta \sum_{j=t+1}^T \delta^{j-t} u_j, \quad (1)$$

where $0 < \beta \leq 1$, $0 < \delta \leq 1$. If $\beta = 1$, the quasi-hyperbolic discounting model reduces to the classical exponential discounting model with discount parameter δ . Compared with exponential discounting, $\beta < 1$ implies that the immediate utility has a higher relative impact on utility. Lower values of β imply a stronger taste for immediate gratification.

With respect to our work, if $\beta = 1$, which implies time-consistent preferences, project delay is explainable by Parkinson's Law. The motivations for Parkinson's Law that are discussed by Krakowski (1974) and summarized in Section 1 can be represented using δ alone. However, an explanation of project delay that includes present bias requires the use of both δ and β . Our work makes the assumption that task operators suffer from present bias. Furthermore, they are *naive*, which means that they understand their immediate bias but not its effect on their future decision making. Depending on how the task operators view their future present bias and how they balance current and future decisions, they can also be modeled as partially naive or sophisticated (O'Donoghue and Rabin 2001). As a first step toward incorporating time-inconsistent behavior into project management, we focus on the naive task operator in this paper. For the sophisticated and partially naive cases, we provided a short discussion and preliminary formulation in Appendix B.

2.1. Basic Setting

The task operator is required to complete a task with workload W , which is a positive integer. In period t , the task operator chooses to spend effort level e_t on the task. We assume that e_t takes only positive integer values, that is, $e_t \in \{1, 2, \dots, e^u\}$, where e^u is the maximum effort level in a period. We let Q_t denote the uncertain productivity during period t , where $\{Q_1, Q_2, \dots\}$ are independent and identically distributed positive discrete random variables with realizations $\{q_1, \dots, q_n\}$ and respective probabilities $\{p_1, \dots, p_n\}$. Then, effort level e_t generates a random amount of completed work $Q_t e_t$ in period t . We let S_t denote the unfinished amount of work at the beginning of period t , where $S_1 = W$. The dynamics of S_t satisfy

$$S_{t+1} = \max \{S_t - Q_t e_t, 0\}, \quad t = 1, \dots$$

From the above definitions, the completion time of the task is

$$\tau = \min \{t \mid S_{t+1} = 0\}.$$

The deadline of the task, defined by a project planning decision, is predetermined at D periods. We assume that the task operator's utility contains two components. The first, based on a preference for leisure over work, is the working-hour-disutility $u_1(\cdot)$ throughout task execution. The second, based on avoidance of blame for late delivery, is the unfinished-task-disutility $u_2(\cdot)$ after the deadline. Both $u_1(\cdot)$ and $u_2(\cdot)$ are

increasing convex functions. We assume that the task operator is a naive decision maker with present bias, whose time preference is a quasi-hyperbolic discounting function with parameters (β, δ) .

At the start of period t ($< D$), the task operator faces the following decision problem

$$(P_t) \min_{e_t, \dots, e_\tau} E_{t, S_t} \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma \beta \sum_{k=D}^{\tau} \delta^{k-t} u_2(S_{k+1}) \right] \\ \text{s.t. } S_{j+1} = \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \\ j = t, \dots, \tau,$$

where $E_{t, S_t}[\cdot] = E[\cdot | S_t] = E[\cdot | Q_1, \dots, Q_{t-1}]$ and γ is a weighting parameter for the unfinished-task-disutility. At the start of period t ($\geq D$), the task operator faces the following decision problem

$$(P_t) \min_{e_t, \dots, e_\tau} E_{t, S_t} \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma u_2(S_{t+1}) \right. \\ \left. + \gamma \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_2(S_{k+1}) \right] \\ \text{s.t. } S_{j+1} = \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \\ j = t, \dots, \tau.$$

We denote the optimal effort solution of (P_t) found at time t by $\{e_t^t(S_t), e_{t+1}^t(S_{t+1}), \dots\}$. However, the naive task operator does not follow this solution after time t . From the behavioral economics literature (for example, O'Donoghue and Rabin 1999a), the task operator reconsiders his or her problem at time $t+1$ and applies a new policy $e_{t+1}^{t+1}(S_{t+1})$, and so on. Because of present bias, in general, $e_{t+1}^{t+1}(S_{t+1}) \neq e_{t+1}^t(S_{t+1})$, that is, the decision about period $t+1$ found at period $t+1$ differs from that found at period t , which illustrates time-inconsistent behavior. The naive task operator simply chooses the first element of the available decision sequence at each time instant, that is, $\{e_1^1(S_1), e_2^2(S_2), \dots\}$. These decisions are not in general optimal for problem (P_0) or (P_t) for any $t > 0$. In the following section, we show how to use these decisions to find the distribution of the completion time, τ , of the task.

Both partially naive and sophisticated task operators also have a time-inconsistency issue. Decision procedures for partially naive and sophisticated task operators can be modeled as sequential intrapersonal games among the task operators at different time instants (Grenadier and Wang 2007, Basak and Chabakauri 2010), as discussed in Appendix B. This paper focuses on designing and analyzing incentive schemes to address the present bias of naive task operators who incorrectly believe that they will perform as time-consistent decision makers in the future. We also mention extensions to various levels of naivety as a possible topic for future research in Section 8.

2.2. Solution Scheme

When $q_i = m_i/M$, where m_i and M are positive integers, the possible realizations of S_t must be less than

$S_t^\# := WM + 1 - (t-1)\min_i\{m_i\}$ and the possible maximum time to finish the task is $T := \lfloor \frac{WM + \min_i\{m_i\} - 1}{\min_i\{m_i\}} \rfloor$ periods, where $\lfloor x \rfloor$ denotes the integer part of x . Problem (P_t) can be solved efficiently. We propose an algorithm to derive the actually applied decisions of the naive task operator $\{e_1^1(S_1), e_2^2(S_2), \dots\}$. We first define the following functions,

$$H(e_t; S_t, t) = g(e_t; S_t, t) + \beta \delta E_{t, S_t}[J(S_{t+1}, t+1)], \quad (2)$$

where

$$g(e_t; S_t, t) = \begin{cases} u_1(e_t), & \text{if } t < D, \\ u_1(e_t) + \gamma E_{t, S_t}[u_2(S_{t+1})], & \text{if } t \geq D; \end{cases} \quad (3)$$

and the auxiliary cost-to-go function $J(S_t, t)$ is defined as

$$J(S_t, t) = \min_{e_t \in \{1, 2, \dots, e^u\}} g(e_t; S_t, t) + \delta E_{t, S_t}[J(S_{t+1}, t+1)], \quad (4)$$

where $J(0, t) = 0$. Because the task operator is a naive decision maker, $\beta J(S_t, t) \neq \min_{e_t} H(e_t; S_t, t)$; hence, the decision system of the naive task operator does not satisfy Bellman's principle of optimality. Therefore, we define $J(S_t, t)$ as an auxiliary cost-to-go function, which is independent of β . Intuitively, because the naive task operator naively believes that his or her future self will behave as a time-consistent operator, the auxiliary cost-to-go function is exactly the cost-to-go function of a time-consistent task operator.

Algorithm 1, which follows, computes the decisions made by a naive task operator, $\{e_t^t(S_t)\}$, and stores them in an $S_1^\# \times T$ table.

Algorithm 1 (Procedure Decisions)

1. At period $T+1$, we set $J(0, T+1) = 0$.
2. At period $t \leq T$, for $S_t \in \{0, 1/M, \dots, S_t^\#/M\}$, we compute $e_t^t(S_t)$ according to

$$e_t^t(S_t) = \underset{e_t \in \{1, 2, \dots, e^u\}}{\operatorname{argmin}} H(e_t; S_t, t),$$

and compute the corresponding auxiliary cost-to-go function $J(S_t, t)$.

3. While $t > 1$, decrease the time t by 1 period, and return to Step 2.
4. When $t = 1$, stop.

There are four nested loops in Algorithm 1. The first layer is for the periods, which executes T times. The second layer is for the state, which executes at most $S_1^\#$ times. The third layer is for the computations of $e_t^t(S_t)$ and $J(S_t, t)$, which executes $2e^u$ times. The fourth layer is for the computation of $E_{t, S_t}[J(S_{t+1}, t+1)]$, which executes n times. Thus, the computational complexity of Algorithm 1 is $O(ne^u TS_1^\#)$.

After obtaining the decision table from Algorithm 1, we derive the distribution of τ using the following algorithm.

Algorithm 2 (Makespan)

1. Generate n^T paths for $\{Q_j\}_{j=1}^T$. The path is denoted as $\{Q_j^R\}_{j=1}^T$, where $Q_j^R \in \{q_1, \dots, q_n\}$ is a realization of Q_j . The probability of this path is $\prod_{j=1}^T Pr(Q_j = Q_j^R)$, where $Pr(Q_j = Q_j^R)$ denotes the probability of $Q_j = Q_j^R$.
2. For each path $\{Q_j^R\}_{j=1}^T$, compute the value of τ using the following steps:
 - i. set $S_1 = W$,
 - ii. at time t , choose $e_t^t(S_t)$ from the decision table and compute $S_{t+1} = S_t - Q_j^R e_t^t(S_t)$,
 - iii. while $S_{t+1} > 0$, increase t by one period and go back to step ii.
 - iv. when $S_{t+1} = 0$, set $\tau(\{Q_j^R\}_{j=1}^T) = t$, and stop.
3. Construct the distribution of τ , which has the realization $\tau(\{Q_j^R\}_{j=1}^T)$ with probability $\prod_{j=1}^T Pr(Q_j = Q_j^R)$.

If n^T is large, we estimate the distribution of τ by simulating M paths of $\{Q_j\}_{j=1}^T$. There are two nested loops in Algorithm 2. The first layer is for the number of simulations, which executes M times. The second layer is for the periods, which executes at most T times. Thus, the computational complexity of Algorithm 2 is $O(MT)$.

In the above model, present bias causes the task operator to discount the cost or benefit that occurs even one period ahead by an additional parameter β . This prioritizes short-term over long-term utility maximization. Also, the naivety of the task operator induces him or her to deviate from the planned decision path repeatedly, which causes further loss of long-term utility. As we discuss in the next section, these features play an *additional* role in project delay that is not explainable by Parkinson's Law. In Section 4, we design an incentive scheme to target these two features.

3. The Impact of Present Bias

As evidenced by the substantial literature discussed in Section 1 about delays that occur in projects, there are a number of potential causes of such delays. Without knowledge of task operators' conscious or even subconscious decision-making processes, fully isolating the effect of present bias requires a controlled experiment, which is beyond the scope of our paper. In this section, we use simulation to isolate the impact of present bias. We apply well-documented estimates of the present bias coefficient β into the decision model of Section 2 and observe the resulting distribution of project completion times. By comparing statistical measures of this simulated distribution to those from empirically observed data about project management performance, we identify the impact of β on project performance. Our simulation results show that present bias is strongly influential on project performance, which motivates its

Table 1. Empirical Estimates of β from the Behavioral Economics Literature

References	Estimate of β
Paserman (2008)	0.40
Della Vigna et al. (2017)	0.58
Ericson (2017)	0.61
Paserman (2008)	0.65
Meier and Sprenger (2015)	0.78
Kaur et al. (2015)	0.82
Paserman (2008)	0.89

use in the incentive scheme, WMT, which we propose in Section 4.

3.1. Present Bias and Project Delay

Several studies within the behavioral economics literature empirically estimate values of β . From these estimates, summarized in Table 1, we conclude that β has an average value of 0.65 and the empirical range [0.40, 0.89] includes six standard deviations. Those studies also estimate the time discount factor, δ . Meier and Sprenger (2015) estimate $\delta = 0.954$ on a monthly basis or $\delta \approx 0.9883$ for a weekly period. Della Vigna et al. (2017) estimate $\delta = 0.995$ on a two-week basis or $\delta \approx 0.9975$ for a weekly period. Paserman (2008) estimates $\delta = 0.9988$ for a weekly period. The mean of these estimates gives $\delta = 0.995$ for a weekly period, the value we use in our computational study.

Empirical evidence about the distribution of project completion times appears in Figure 1, which is figure 4 of Lewis and Bajari (2014, p.1212), based on 466 construction projects in Minnesota. We observe a concentration of completion times around the project deadline. Also, the remainder of the distribution is asymmetric, with a fat right tail representing project delay. We measure those features using skewness and kurtosis. For a normal distribution, the skewness is zero and the kurtosis

Figure 1. (Color online) Empirically Observed Project Completion Times

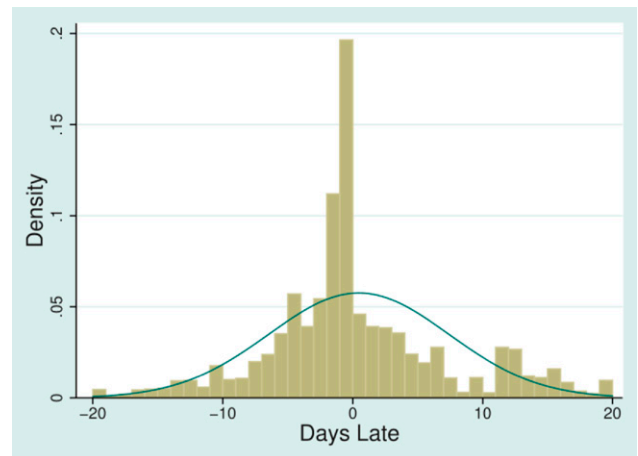
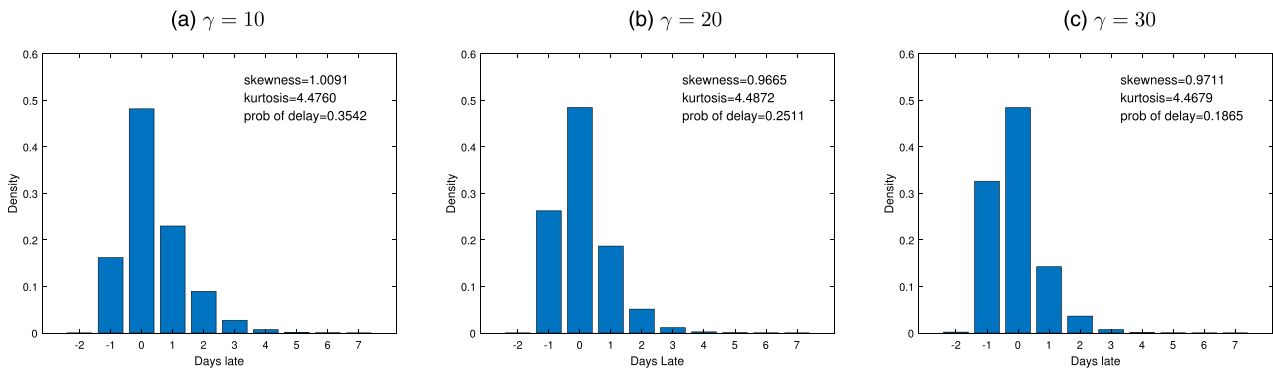


Figure 2. (Color online) Distribution of Task Completion Times with $\beta = 1$ 

is three. In Figure 1, the skewness is 4.0647 and the kurtosis is 34.0836. We also consider the probability of project delay, which is 0.3283 in Figure 1. Below, we show that β is important in generating large skewness, kurtosis, and probability of delay.

3.2. Simulation Results at Task Level

As a benchmark example, we set $c^u = 12$, $q_i = i/4$ ($i = 1, \dots, 7$), $p_i = 1/7$ ($i = 1, \dots, 7$), $u_1(x) = x^2$ (convex), $u_2(x) = x^2$ (convex), $W = 48$, $\delta = 0.995$, and $D = 6$. Without considering any behavioral parameters, the task operator minimizes effort, and the project completion time distribution is close to normal. Now, we set the weighting parameter for unfinished-task-disutility to be $\gamma \in \{10, 20, 30\}$. The distribution of task completion times is shown in Figure 2. The addition of the unfinished-task-disutility increases the probability of project completion around the deadline. A larger parameter γ motivates the task operator to work harder around the deadline, which generates a smaller probability of delay and a smaller value of skewness. Also, postponement by a task operator with severe present bias has a similar effect on task performance to slow work by a task operator with weak incentive to avoid delay.

To separate the effects of β and γ , we fix the unfinished-task-disutility parameter at $\gamma = 30$ and let $\beta \in$

$\{0.9, 0.6, 0.3\}$ (see Figure 3). Here, the effect of decreasing the value of β is striking. As β decreases from 0.9 to 0.3, the distribution becomes strongly asymmetric, with skewness increasing from 0.9867 to 1.5218 and kurtosis increasing from 4.7621 to 5.9583. Also, the probability of delay increases from 0.2041 to 0.3749.

Figure 4 further isolates the effect of present bias from that of unfinished-task-disutility. For all values of $\gamma \geq 5$, the impact of present bias is considerable. When the task operator has the mean empirical value $\beta = 0.65$, both skewness and kurtosis reach their highest values 1.294 and 4.958 at $\gamma = 10$. When $\beta = 0.4$, skewness attains its highest value 1.404 at $\gamma = 19$, where more than one-third of the empirically observed skewness can be explained. Also, kurtosis attains its highest value 5.583 at $\gamma = 28$, where more than one-sixth of the empirically observed kurtosis can be explained. As β decreases to 0.4, the probability of delay increases monotonically from 0.35 to 0.54 for $\gamma = 10$ and increases monotonically from 0.09 to 0.19 for $\gamma = 100$.

Figure 5 further shows how present bias directly influences the probability of delay. For any choice of γ , changing β from one to zero increases the probability of delay to one. To summarize, present bias plays a significant role in generating large skewness and kurtosis

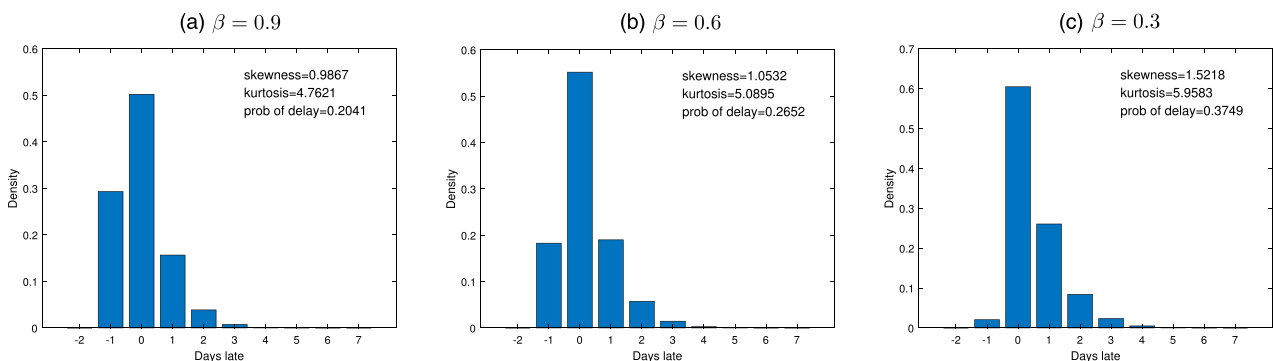
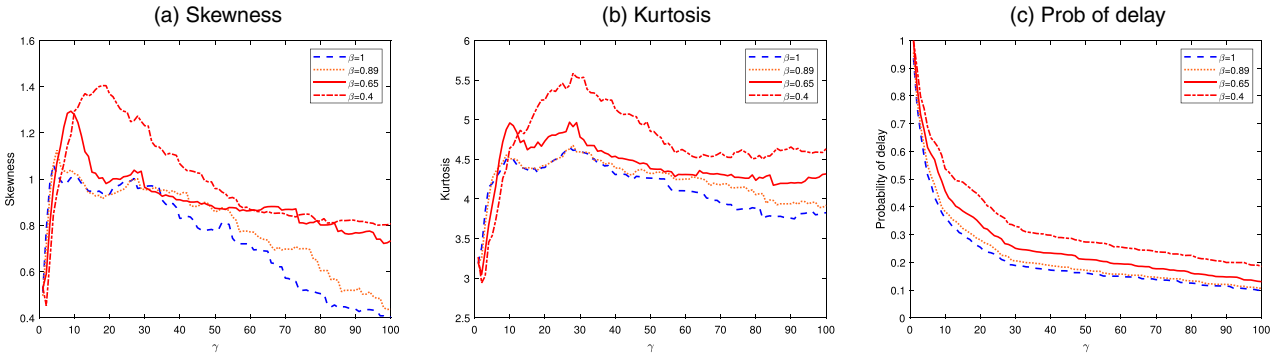
Figure 3. (Color online) Distribution of Task Completion Times with $\gamma = 30$ 

Figure 4. (Color online) Skewness, Kurtosis, and Probability of Delay Under Various Parameter Settings



of project completion times and large probability of delay.

3.3. Simulation Results at Project Level

Consider a serial network with N tasks, generated as in Section 3.2. For the task operators, we set $\gamma = 10$ and change β from zero to one. We assume that the finishing times of different tasks are independent and allow the project manager to adjust the allowed task time for the remaining tasks based on the previous task performance (see Section 6.1).

Figure 6 shows that reducing β generates much larger skewness and kurtosis than at the task level, for example, 1.847 and 8.33, respectively, when $n = 120$. Importantly, the largest skewness and kurtosis are both attained around $\beta = 0.62$, which is very close to the average estimated value of $\beta = 0.65$ in the literature. Also, β has a large impact on the probability of project delay. Overall, the effect of present bias on skewness, kurtosis, and probability of delay is much larger at the project level and more similar to the empirically observed data than at the task level.

4. Incentive Design

The purpose of incentives is to improve project performance. We define project performance as (a) the probability of the project being completed on time, $Pr(\tau \leq D)$, and (b) the expected tardiness of the project, $E[(\tau - D)1_{\{\tau > D\}}]$, where $1_{\{\cdot\}}$ is the indicator function. Both are standard measures of time performance in projects (Bertsimas et al. 2004, Kerzner 2017), motivated in many cases by contract penalties. Besides these two performance measures, which directly assess the outcome of different incentive schemes, we also consider the cost of applying different incentive schemes. A better incentive scheme should achieve better performance with the same level of cost or the same performance with less cost.

We propose an incentive design scheme inspired by the Save More Tomorrow (SMT) program (Thaler and Benartzi 2004). In the SMT program, people commit in advance to allocate a portion of their future salary increases toward retirement savings. The key feature “commit in advance” targets the time-inconsistent behavior in saving that results from present bias. By drawing an analogy between the cost of savings experienced by an employee and the effort required by a task operator to complete a task, we propose a Work More Tomorrow scheme to target the delay in project execution that is caused by present bias. The main idea behind WMT is that if the operator is already late today from postponing an unpleasant task due to present bias, then he or she should carry the unfinished work into tomorrow’s workplan. Then, a bonus or penalty motivates the operator with present bias to work more in the next period, making on-time delivery of the project more likely.

Figure 5. (Color online) Probability of Delay as a Function of β

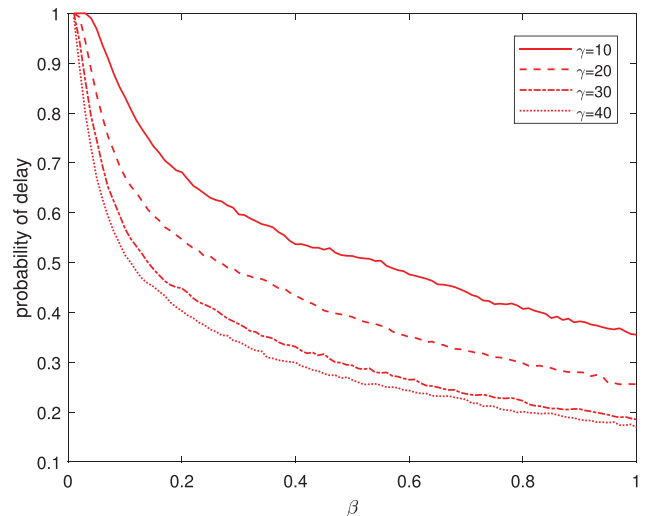
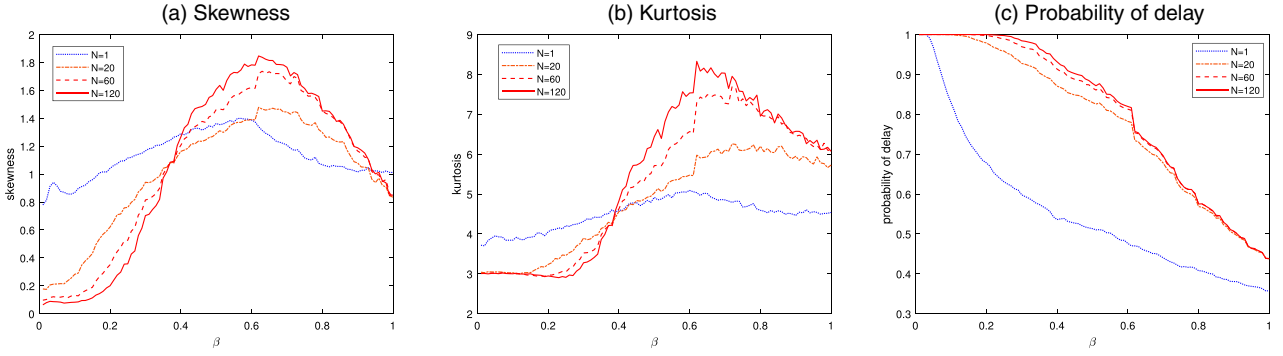


Figure 6. (Color online) Skewness, Kurtosis, and Probability of Delay Under Different Numbers of Tasks

Because the project manager typically observes the realized progress S_t but not the effort level e_t of the task operator, we propose the following WMT scheme based on the task progress.

1. *Target setting*: The plan specifies that the task operator will complete workload S_1/D in each period to complete the task on time. (i) When the task progress is satisfactory up to period t , that is, $S_t \leq S_1(D-t+1)/D$, the plan specifies that the task operator will complete the regular workload S_1/D in period t and thereby achieve the task progress target $\max\{S_t - S_1/D, 0\}$. (ii) When the task progress is behind schedule up to period t , that is, $S_t > S_1(D-t+1)/D$, the plan specifies that the task operator will complete the regular workload S_1/D and the delayed workload $[S_t - S_1(D-t+1)/D]$ in period t and thereby achieve the task progress target $S_1(D-t)/D$. The task progress target at the end of period t is thus,

$$S_{t+1}^{\text{target}} = \begin{cases} \max\left\{S_t - \frac{S_1}{D}, 0\right\}, & \text{if } S_t \leq \frac{S_1(D-t+1)}{D} \text{ and } t < D, \\ \frac{S_1(D-t)}{D}, & \text{if } S_t > \frac{S_1(D-t+1)}{D} \text{ and } t < D, \\ 0, & \text{if } t \geq D. \end{cases} \quad (5)$$

If the operator is late, for example, at period 1, then the unfulfilled workload is added into the period 2 progress target.

2. *Progress control*: At period t , the task operator can obtain a bonus or penalty based on whether the task progress target is reached or not, which is computed as

$$I_t = \begin{cases} (b1_{\{S_{t+1}^{\text{target}} > S_{t+1}\}} + p1_{\{S_{t+1}^{\text{target}} \leq S_{t+1}\}})(S_{t+1}^{\text{target}} - S_{t+1}), & \text{if } t < D, \\ -pS_{t+1}, & \text{if } t \geq D, \end{cases} \quad (6)$$

where $b > 0$ and $p > 0$ are a bonus and penalty per unit

of workload, respectively. The cumulative incentive until period t is

$$NI_{t+1} = \sum_{j=1}^t I_j.$$

3. *Final settlement*: When the project is finished on time and the cumulative incentive is negative (corresponding to a net penalty), the penalty is eliminated. When the project is delayed and the cumulative incentive is positive (corresponding to a net bonus), the bonus is eliminated. To implement this, the task operator incurs a final settlement at time $\min\{\tau, D\}$ as follows:

$$-1_{\{\tau \leq D\}} 1_{\{NI_\tau < 0\}} NI_\tau - 1_{\{S_{D+1} > 0\}} 1_{\{NI_D > 0\}} NI_D.$$

As a benchmark incentive scheme, we consider a linear contract (LC), which is now defined. The linear contract is a bonus for early completion and a penalty for late completion

$$(B1_{\{\tau < D\}} + P1_{\{\tau \geq D\}})(D - \tau),$$

where $B > 0$ and $p > 0$ are a bonus and penalty per period, respectively. This scheme is apparently the most widely used in practice and discussed in the project management literature (Abu-Hilje and Ibbs 1989, Arditi and Yasamis 1998, Bower et al. 2002, Bayiz and Corbett 2005). For the benchmark scheme, the bonus or penalty is settled at the end of the task, which is typically several periods into the future. This is in contrast with the WMT scheme, where the bonus or penalty is accrued each period, or because of administrative constraints, every few periods. However, for purposes of fair comparison we also allow period-by-period settlement of the benchmark scheme (see Section 5.3).

Sections 5, 6, and 7 compare the performance of the WMT and LC schemes for a single task, a serial network, and a general acyclic network, respectively.

5. Single Task

In this section, we study the performance of the WMT and LC incentive schemes for a single task.

Section 5.1 presents a theoretical comparison between the performance of the two schemes. Section 5.2 provides a computational comparison. We begin by formally defining the decision problems under both schemes.

The decision problem under the WMT scheme (P_t^{WMT}) is formulated as follows.

When $t < D$,

$$\begin{aligned} \min_{e_t, \dots, e_\tau} E_{t, S_t, NI_t} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) \right. \\ & + \gamma \beta \sum_{k=D}^{\tau} \delta^{k-t} u_2(S_{k+1}) - I_t - \beta \sum_{k=t+1}^{\tau} \delta^{k-t} I_k \\ & + 1_{\{S_{t+1}=0\}} 1_{\{NI_t < 0\}} NI_t + \beta \sum_{k=t+1}^D \delta^{k-t} 1_{\{S_{k+1}=0\}} 1_{\{NI_k < 0\}} NI_k \\ & \left. + \beta \delta^{D-t} 1_{\{S_{D+1} > 0\}} 1_{\{NI_D > 0\}} NI_D \right] \\ \text{s.t. } S_{j+1} &= \max \{S_j - Q_j e_j, 0\}, \quad NI_{j+1} = NI_j + I_j, \\ e_j &\in \{1, 2, \dots, e^u\}, \quad j = t, \dots, \tau. \end{aligned}$$

When $t = D$,

$$\begin{aligned} \min_{e_t, \dots, e_\tau} E_{t, S_t, NI_t} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma u_2(S_{t+1}) \right. \\ & + \gamma \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_2(S_{k+1}) - I_t - \beta \sum_{k=t+1}^{\tau} \delta^{k-t} I_k \\ & \left. + 1_{\{S_{t+1}=0\}} 1_{\{NI_t < 0\}} NI_t + 1_{\{S_{t+1} > 0\}} 1_{\{NI_t > 0\}} NI_t \right] \\ \text{s.t. } S_{j+1} &= \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \\ j &= t, \dots, \tau. \end{aligned}$$

When $t > D$,

$$\begin{aligned} \min_{e_t, \dots, e_\tau} E_{t, S_t} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma u_2(S_{t+1}) \right. \\ & + \gamma \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_2(S_{k+1}) - I_t - \beta \sum_{k=t+1}^{\tau} \delta^{k-t} I_k \\ & \left. + 1_{\{S_{t+1}=0\}} 1_{\{NI_t < 0\}} NI_t + 1_{\{S_{t+1} > 0\}} 1_{\{NI_t > 0\}} NI_t \right] \\ \text{s.t. } S_{j+1} &= \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \\ j &= t, \dots, \tau, \end{aligned}$$

where the task progress target S_{t+1}^{target} is computed from (5) and the bonus or penalty at period t , I_t , is computed from (6). The decision problem under the benchmark LC scheme (P_t^{LC}) is as follows.

When $t < D$:

$$\begin{aligned} \min_{e_t, \dots, e_\tau} E_{t, S_t} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma \beta \sum_{k=D}^{\tau} \delta^{k-t} u_2(S_{k+1}) \right. \\ & - B(D-t) 1_{\{\tau=t\}} \\ & \left. - \beta \sum_{k=t+1}^T \delta^{k-t} (B 1_{\{k < D\}} + P 1_{\{k \geq D\}}) (D-k) 1_{\{\tau=k\}} \right] \\ \text{s.t. } S_{j+1} &= \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \\ j &= t, \dots, \tau. \end{aligned}$$

When $t \geq D$,

$$\begin{aligned} \min_{e_t, \dots, e_\tau} E_{t, S_t} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma u_2(S_{t+1}) \right. \\ & + \gamma \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_2(S_{k+1}) - P(D-t) 1_{\{\tau=t\}} \\ & \left. - \beta \sum_{k=t+1}^T \delta^{k-t} P 1_{\{k \geq D\}} (D-k) 1_{\{\tau=k\}} \right] \\ \text{s.t. } S_{j+1} &= \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \\ j &= t, \dots, \tau. \end{aligned}$$

By adding the state variable NI_t , the decisions actually applied by the naive task operator under the WMT scheme, $e_t^{WMT}(S_t, NI_t)$, and the completion time of the task under the WMT scheme can be derived from Algorithm 1 and Algorithm 2, with $H(e_t; S_t, t)$ in (2) and $J(S_t, t)$ in (4), respectively, replaced by

$$\begin{aligned} H_{WMT}(e_t; S_t, NI_t, t) &= g(e_t; S_t, t) + E_{t, S_t, NI_t} [h_{WMT}(S_{t+1}; S_t, NI_t, t)] \\ &\quad + \beta \delta E_{t, S_t, NI_t} [J_{WMT}(S_{t+1}, NI_{t+1}, t+1)], \\ J_{WMT}(S_t, NI_t, t) &= \min_{e_t \in \{1, 2, \dots, e^u\}} \{g(e_t; S_t, t) + E_{t, S_t, NI_t} [h_{WMT}(S_{t+1}; S_t, NI_t, t)] \\ &\quad + \delta E_{t, S_t, NI_t} [J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]\}, \end{aligned} \quad (7)$$

where $J_{WMT}(0, NI_t, t) = 0$, and

$$\begin{aligned} h_{WMT}(S_{t+1}; S_t, NI_t, t) &= \begin{cases} -I_t + 1_{\{S_{t+1}=0\}} 1_{\{NI_t < 0\}} NI_t, & t < D, \\ -I_t + 1_{\{S_{t+1}=0\}} 1_{\{NI_t < 0\}} NI_t + 1_{\{S_{t+1} > 0\}} 1_{\{NI_t > 0\}} NI_t, & t = D, \\ -I_t, & t > D. \end{cases} \end{aligned}$$

As Algorithm 2 for the WMT scheme only depends on the periods T and simulation numbers M , its computational complexity is still $O(MT)$. However, because the new state variable NI_t is added into the state space for the periods $t < D$, the computational complexity of Algorithm 1 becomes larger. The largest and smallest workloads completed in a period are $\max_i \{m_i\} e^u / M$ and $\min_i \{m_i\} / M$, respectively. When $b = p$, the number of possible realizations of NI_{t+1} ($t < D$) is

$$\begin{aligned} NI_{t+1}^\# &= \left(\frac{W}{D} - \frac{\min_i \{m_i\}}{M} \right) \frac{t(t+1)M}{2} \\ &\quad + \min \left\{ \frac{W(D-t)}{D}, \frac{\max_i \{m_i\} e^u t}{M} - \frac{Wt}{D} \right\} M + 1, \end{aligned}$$

where the first term denotes the number of possible negative realizations and the second term denotes the number of possible positive realizations. We denote

the largest value of $NI_{t+1}^\#$ over $t = 1, \dots, D-1$ as $NI^\#$. As the state variable NI_t only exists for the periods $t < D$, the computational complexity for $t < D$ is $O(ne^u DS_1^\# NI^\#)$. Also, the computational complexity for $t \geq D$ is $O(ne^u (T-D) S_1^\#)$. Thus, the computational complexity of Algorithm 1 is $O(\max\{D \cdot NI^\#, (T-D)\} \cdot ne^u S_1^\#)$.

Meanwhile, the decisions actually applied by the naive task operator under the LC scheme, $e_t^{LC}(S_t)$, and the completion time of the task under the LC scheme can be derived by Algorithm 1 and Algorithm 2 with $H(e_t; S_t, t)$ and $J(S_t, t)$ replaced by

$$\begin{aligned} H_{LC}(e_t; S_t, t) &= g(e_t; S_t, t) + E_{t,S_t}[h_{LC}(S_{t+1}; S_t, t)] \\ &\quad + \beta \delta E_{t,S_t}[J_{LC}(S_{t+1}, t+1)], \\ J_{LC}(S_t, t) &= \min_{e_t \in \{1, 2, \dots, e^u\}} \{g(e_t; S_t, t) + E_{t,S_t}[h_{LC}(S_{t+1}; S_t, t)] \\ &\quad + \delta E_{t,S_t}[J_{LC}(S_{t+1}, t+1)]\}, \end{aligned} \quad (8)$$

where $J_{LC}(0, t) = 0$ and

$$h_{LC}(S_{t+1}; S_t, t) = \begin{cases} -B(D-t)1_{\{S_{t+1}=0\}}1_{\{S_t>0\}} & \text{if } t < D, \\ P(t-D)1_{\{S_{t+1}=0\}}1_{\{S_t>0\}} & \text{if } t \geq D. \end{cases}$$

Because the state variables are still S_t and t , the computational complexities of Algorithm 1 and Algorithm 2 for the LC scheme are also $O(ne^u TS_1^\#)$ and $O(MT)$, respectively.

5.1. Theoretical Comparison

We compare the WMT and LC schemes theoretically for a single task. We compare the objective functions under the two schemes. The objective function of WMT, from (7), has three different terms. The first term $g(e_t; S_t, t)$ is the cost of effort, which is not related to the incentive scheme. The remaining terms $E_{t,S_t,NI_t}[h_{WMT}(S_{t+1}; S_t, NI_t, t)]$ and $\beta \delta E_{t,S_t,NI_t}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]$ are directly related to the incentive scheme and represent the current penalty and future penalty terms. The objective function of LC, from (8), also has three similar terms. Although the first terms are the same in the WMT and LC schemes, the second and third terms are different and play an important role in motivating the task operator differently. We provide a lemma to describe the properties of these terms under both the WMT and LC schemes. For simplicity, but without loss of generality, we set $u_1(0) = 0$, $\min_i \{m_i\} = 1$ and $\delta = 1$.¹

Lemma 1.

a. For the WMT scheme, its current penalty function $h_{WMT}(S_{t+1}; S_t, NI_t, t)$ is a strictly increasing function of S_{t+1} , and $E_{t,S_t,NI_t}[h_{WMT}(S_{t+1}; S_t, NI_t, t)]$ is a nonincreasing function

of e_t ; the future penalty function $J_{WMT}(S_{t+1}, NI_{t+1}, t+1)$ is a nondecreasing function of NI_{t+1} ; when $t+1 > D$, $J_{WMT}(S_{t+1}, NI_{t+1}, t+1)$ is an increasing function of S_{t+1} and $E_{t,S_t,NI_t}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]$ is a nonincreasing function of e_t .

b. For the LC scheme, its future penalty function $J_{LC}(S_{t+1}, t+1)$ is an increasing function of S_{t+1} (for $S_{t+1} > 0$) and $J_{LC}(S_{t+1}, t+1) > h_{LC}(0; 1/M, t)$, $E_{t,S_t}[J_{LC}(S_{t+1}, t+1)]$ is a nonincreasing function of e_t . The current penalty function has more complicated properties:

Case I: When $t < D$ and $S_t \leq \max_i \{q_i e^u\}$ (i.e., the task is on schedule), the current penalty function $h_{LC}(S_{t+1}; S_t, t)$ is a nondecreasing step function of S_{t+1} and $E_{t,S_t}[h_{LC}(S_{t+1}; S_t, t)]$ is a nonincreasing function of e_t .

Case II: When $t < D$ and $S_t > \max_i \{q_i e^u\}$ (i.e., the task is not yet delayed but the remaining workload is large), the current penalty function $h_{LC}(S_{t+1}; S_t, t)$ is a nondecreasing step function of S_{t+1} and $E_{t,S_t}[h_{LC}(S_{t+1}; S_t, t)]$ is a constant function of e_t .

Case III: When $t \geq D$ and $S_t > 0$ (i.e., the task is already overdue), the current penalty function $h_{LC}(S_{t+1}; S_t, t)$ is a nonincreasing step function of S_{t+1} and $E_{t,S_t}[h_{LC}(S_{t+1}; S_t, t)]$ is a nondecreasing function of e_t .

Lemma 1 describes how well different incentive schemes motivate the task operator under different conditions. Observe that, to be an efficient incentive scheme, the penalty function should be a nondecreasing function of S (i.e., greater unfinished workload leads to greater penalty) and a nonincreasing function of e (i.e., larger effort leads to smaller penalty). Lemma 1(a) shows that the WMT scheme works efficiently: both the current and future penalty functions are increasing in S and nonincreasing in e . The LC scheme performs well under Case I but works inefficiently under Cases II and III. From Lemma 1(b), the future penalty function of the LC scheme is always a nonincreasing function of e_t , which encourages harder work. However, for Case III, with $t > D$, the current penalty term of the LC scheme is a nondecreasing function of e_t , which does not encourage harder work; for Case II, with $t < D$ and $S_t > \max_i \{q_i e^u\}$, the current penalty term of LC scheme is a constant function of e_t , which does not encourage harder work either. When β is large (recall that the future penalty term is discounted by $\beta \delta$), the future penalty term dominates and the LC scheme is an effective incentive scheme to motivate harder work. When β is small, however, the current penalty term dominates; hence, the LC scheme performs inefficiently under Cases II and III.

To summarize, the LC scheme performs well under Case I but works inefficiently under Cases II and III. However, severe present bias of the task operators (shown by a small β value) drives the project, especially a complicated project with many tasks, into Case III and Case II scenarios, where either the project is

already overdue or the project is not yet delayed but the remaining workload accumulates to a high level. For those two scenarios, the WMT scheme performs better theoretically than the LC scheme.

In the following theorem, we provide necessary and sufficient conditions under which the WMT scheme motivates a larger effort level than the LC scheme under all three cases. Given t , S_t , and NI_t , we can simplify the following expressions:

$$\begin{aligned}\tilde{g}(e_t) &= g(e_t; S_t, t), \quad \tilde{h}_L(e_t) = E_{t, S_t} [h_{LC}(S_{t+1}; S_t, t)], \\ \tilde{J}_L(e_t) &= E_{t, S_t} [J_{LC}(S_{t+1}, t + 1)], \\ \tilde{h}_W(e_t) &= E_{t, S_t, NI_t} [h_{WMT}(S_{t+1}; S_t, NI_t, t)], \\ \tilde{J}_W(e_t) &= E_{t, S_t, NI_t} [J_{WMT}(S_{t+1}, NI_{t+1}, t + 1)], \\ \Delta \tilde{g}(i, e_t) &= \tilde{g}(i) - \tilde{g}(e_t), \quad \Delta \tilde{h}_L(i, e_t) = \tilde{h}_L(i) - \tilde{h}_L(e_t), \\ \Delta \tilde{J}_L(i, e_t) &= \tilde{J}_L(i) - \tilde{J}_L(e_t), \quad \Delta \tilde{h}_W(i, e_t) = \tilde{h}_W(i) - \tilde{h}_W(e_t), \\ \Delta \tilde{J}_W(i, e_t) &= \tilde{J}_W(i) - \tilde{J}_W(e_t).\end{aligned}$$

Theorem 1. For the single task case, we have the following results to compare the WMT and LC schemes.

a. Case I: When $t < D$ and $S_t \leq \max_i \{q_i e^u\}$, the necessary and sufficient condition for $e_t^{WMT}(S_t, NI_t) \geq e_t^{LC}(S_t)$ is

$$\beta \in \cup_{i=1}^{e^u} [\underline{\beta}(i), \bar{\beta}(i)],$$

where $\cup$ denotes the union of the intervals and

$$\begin{aligned}\underline{\beta}(i) &= \max \left\{ \max_{1 \leq e_t < i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)} 1_{\{\Delta \tilde{J}_W(e_t, i) > 0\}} \right. \right. \\ &\quad \left. \left. + 1.1 \times 1_{\{\Delta \tilde{J}_W(e_t, i) = 0, \Delta \tilde{g}(i, e_t) > \Delta \tilde{h}_W(e_t, i)\}} \right\}, 0 \right\}, \\ \bar{\beta}(i) &= \min \left\{ \min_{1 \leq e_t < i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)} 1_{\{\Delta \tilde{J}_W(e_t, i) < 0\}} \right. \right. \\ &\quad \left. \left. + 1_{\{\Delta \tilde{J}_W(e_t, i) \geq 0\}} \right\}, \right. \\ &\quad \left. \min_{e^u \geq e_t > i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_L(i, e_t)}{\delta \Delta \tilde{J}_L(e_t, i)} 1_{\{\Delta \tilde{J}_L(e_t, i) < 0\}} \right. \right. \\ &\quad \left. \left. + \left(1 - 1.1 \times 1_{\{\Delta \tilde{g}(i, e_t) > \Delta \tilde{h}_L(e_t, i)\}} \right) 1_{\{\Delta \tilde{J}_L(e_t, i) = 0\}} \right\}, 1 \right\}.\end{aligned}$$

b. Case II: When $t < D$ and $S_t > \max_i \{q_i e^u\}$, the necessary and sufficient condition for $e_t^{WMT}(S_t, NI_t) \geq e_t^{LC}(S_t)$ is

$$\beta \in \cup_{i=1}^{e^u} [\underline{\beta}(i), \tilde{\beta}(i)],$$

where

$$\begin{aligned}\underline{\beta}(i) &= \max \left\{ \max_{1 \leq e_t < i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)} 1_{\{\Delta \tilde{J}_W(e_t, i) > 0\}} \right. \right. \\ &\quad \left. \left. + 1.1 \times 1_{\{\Delta \tilde{J}_W(e_t, i) = 0, \Delta \tilde{g}(i, e_t) > \Delta \tilde{h}_W(e_t, i)\}} \right\}, 0 \right\}, \\ \tilde{\beta}(i) &= \min \left\{ \min_{1 \leq e_t < i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)} 1_{\{\Delta \tilde{J}_W(e_t, i) < 0\}} \right. \right. \\ &\quad \left. \left. + 1_{\{\Delta \tilde{J}_W(e_t, i) \geq 0\}} \right\}, \right. \\ &\quad \left. \min_{e^u \geq e_t > i} \left\{ \frac{\Delta \tilde{g}(i, e_t)}{\delta \Delta \tilde{J}_L(e_t, i)} 1_{\{\Delta \tilde{J}_L(e_t, i) < 0\}} \right. \right. \\ &\quad \left. \left. + \left(1 - 1.1 \times 1_{\{\Delta \tilde{g}(i, e_t) > \Delta \tilde{h}_L(e_t, i)\}} \right) 1_{\{\Delta \tilde{J}_L(e_t, i) = 0\}} \right\}, 1 \right\}.\end{aligned}$$

c. Case III: When $t \geq D$ and $S_t > 0$, the necessary and sufficient condition for $e_t^{WMT}(S_t, NI_t) \geq e_t^{LC}(S_t)$ is

$$\beta \in \cup_{i=1}^{e^u} [\underline{\beta}(i), \bar{\beta}(i)], \text{ where}$$

$$\begin{aligned}\underline{\beta}(i) &= \max \left\{ \max_{1 \leq e_t < i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)} 1_{\{\Delta \tilde{J}_W(e_t, i) > 0\}} \right. \right. \\ &\quad \left. \left. + 1.1 \times 1_{\{\Delta \tilde{J}_W(e_t, i) = 0, \Delta \tilde{g}(i, e_t) > \Delta \tilde{h}_W(e_t, i)\}} \right\}, 0 \right\}, \\ \bar{\beta}(i) &= \min \left\{ \min_{e^u \geq e_t > i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_L(i, e_t)}{\delta \Delta \tilde{J}_L(e_t, i)} 1_{\{\Delta \tilde{J}_L(e_t, i) < 0\}} \right. \right. \\ &\quad \left. \left. + \left(1 - 1.1 \times 1_{\{\Delta \tilde{g}(i, e_t) > \Delta \tilde{h}_L(e_t, i)\}} \right) 1_{\{\Delta \tilde{J}_L(e_t, i) = 0\}} \right\}, 1 \right\}.\end{aligned}$$

Remark 1. We observe that the bounds, $\underline{\beta}(i)$, $\bar{\beta}(i)$, $\tilde{\beta}(i)$, $\tilde{\beta}(i)$, do not depend on the values of β . The reason is that the naive task operator incorrectly believes that his or her future self will perform as a time-consistent decision maker, unaffected by the present bias β . Mathematically, the auxiliary cost-to-go functions $\tilde{J}_L(e_t)$ and $\tilde{J}_W(e_t)$ do not depend on β , which implies that the bounds are also independent of β . However, we want to emphasize that this independence feature only holds for naive task operators. For partially naive and sophisticated task operators, the equivalent conditions under which the WMT scheme motivates a larger effort level than the LC scheme does would depend on β . In Section 8, we mention the analysis of these cases as possible extensions. We also describe a related model in Appendix B.

The expressions for necessary and sufficient conditions on β are complicated, but they are also quite intuitive. The optimal effort under WMT is derived by solving $H_{WMT}(e_t; S_t, NI_t, t) = \tilde{g}(e_t) + \tilde{h}_W(e_t) + \beta \tilde{J}_W(e_t)$ as shown in (7), which is a linear function of β . Therefore,

the optimal effort under WMT is a function of β , denote by $e_t^{WMT}(\beta)$. Similarly, the optimal effort under LC is denoted as $e_t^{LC}(\beta)$. The necessary and sufficient conditions given in Theorem 1 imply that $e_t^{WMT}(\beta) \geq e_t^{LC}(\beta)$.

5.2. Computational Comparison

We illustrate the main results of Theorem 1 by a numerical example.

Example 1. We set $e^u = 12$, $q_i = i/4$ ($i = 1, \dots, 7$), $p_i = 1/7$ ($i = 1, \dots, 7$), $u_1(x) = x^2$, $u_2(x) = x^2$, $W = 48$, $\delta = 0.995$, $D = 6$, and $\gamma = 10$. The β intervals, which identify conditions where the WMT scheme motivates a larger effort level than the LC scheme, depend on the setting of $\{b, p, B, P\}$. In an extreme case, if we set the values of $\{B, P\}$ in LC high enough, the β intervals may be an empty set. To create a fair comparison, we set $b = p = 7$ and $B = P = 56$, which provides the same marginal incentive per period under the two schemes. Under this parameter setting, we can demonstrate that for Case I, $(t, S_t) = (3, 13)$ and $NI_t = 126$, $\cup_{i=1}^{e^u} [\underline{\beta}(i), \bar{\beta}(i)] = [0, 0.5762]$; for Case II, $(t, S_t) = (3, 29)$ and $NI_t = -7$, $\cup_{i=1}^{e^u} [\underline{\beta}(i), \bar{\beta}(i)] = [0, 1]$; and for Case III, $(t, S_t) = (8, 5)$, $\cup_{i=1}^{e^u} [\underline{\beta}(i), \bar{\beta}(i)]$

$= [0, 0.9093]$. We observe that the conditions on β are usually simple intervals starting from zero and the WMT scheme has obvious advantages in Case II and Case III.

The theoretical results in Theorem 1 and Example 1 show that the WMT and LC schemes have relative advantages under different cases. We now compare the performance of WMT and LC by detailed simulation. To ensure a fair comparison between the different schemes, we compute the cost required by the WMT and LC schemes to achieve the same performance target. More specifically, we fix the probability of on-time completion as the target and then compute the accumulated bonus (*bonus*) and the accumulated penalty (*penalty*) required to achieve the given target. Meanwhile, we also record the corresponding expected tardiness (*tardiness*). The other parameters follow the settings in Example 1.

Table 2 reports the results, where the improvement results in the last three columns represent the improvement ratio of the WMT scheme, relative to the LC scheme.² From the results in Table 2, we observe the following:

1. When the task operator has small present bias ($\beta = 0.8$) or no present bias ($\beta = 1$), the LC scheme uses

Table 2. Performance Comparison Between WMT and LC Schemes for a Single Task

β	$Pr(\tau \leq D)$	WMT			LC			Improvement		
		Tardiness	Bonus	Penalty	Tardiness	Bonus	Penalty	Tardiness	Bonus	Penalty
1	0.70	0.454	4.4	2.6	0.427	2.1	3.7	-6.2%	-110.2%	29.3%
	0.75	0.374	12.6	5.9	0.342	6.4	6.4	-9.4%	-97.9%	7.5%
	0.80	0.299	25.5	7.7	0.267	15.5	8.6	-12.0%	-64.6%	10.1%
	0.85	0.226	54.8	10.8	0.197	35.6	10.1	-14.6%	-54.0%	-7.6%
	0.90	0.147	126.1	16.6	0.129	83.7	10.3	-13.6%	-50.7%	-61.8%
0.8	0.70	0.456	11.1	10.3	0.431	6.5	11.5	-5.8%	-71.8%	10.0%
	0.75	0.375	21.9	13.8	0.350	15.7	15.1	-7.1%	-40.1%	8.3%
	0.80	0.302	42.3	16.0	0.278	29.8	15.8	-8.5%	-42.0%	-1.6%
	0.85	0.228	74.9	16.3	0.201	66.2	17.4	-13.6%	-13.1%	6.4%
	0.90	0.148	152.7	21.7	0.133	126.7	15.3	-11.7%	-20.5%	-41.3%
0.6	0.70	0.460	16.4	22.7	0.487	16.6	30.3	5.6%	1.3%	24.9%
	0.75	0.385	30.4	25.1	0.413	37.0	37.5	6.7%	17.9%	33.0%
	0.80	0.308	52.9	25.1	0.322	76.6	40.8	4.2%	30.9%	38.4%
	0.85	0.228	93.7	22.8	0.243	122.3	35.2	6.3%	23.4%	35.3%
	0.90	0.153	175.6	25.0	0.157	206.9	28.2	2.8%	15.1%	11.3%
0.4	0.70	0.464	21.9	37.7	0.697	59.2	115.0	33.4%	62.9%	67.3%
	0.75	0.383	39.8	38.2	0.597	97.8	120.5	35.8%	59.3%	68.3%
	0.80	0.301	69.2	34.9	0.490	152.2	122.4	38.5%	54.6%	71.5%
	0.85	0.222	114.7	31.1	0.371	225.3	105.5	40.2%	49.1%	70.6%
	0.90	0.145	198.8	26.6	0.245	370.0	77.6	40.8%	46.3%	65.7%
0.2	0.70	0.476	29.7	62.4	1.035	213.0	586.9	54.0%	86.0%	89.4%
	0.75	0.387	51.1	56.4	0.866	315.1	558.0	55.4%	83.8%	89.9%
	0.80	0.302	84.0	50.2	0.695	455.5	518.4	56.5%	81.6%	90.3%
	0.85	0.226	136.0	40.4	0.514	608.9	412.6	55.9%	77.7%	90.2%
	0.90	0.151	239.6	39.8	0.331	916.9	261.9	54.3%	73.9%	84.8%
0	0.70	0.476	62.4	142.1	—	—	—	—	—	—
	0.75	0.392	84.9	114.9	—	—	—	—	—	—
	0.80	0.307	117.4	89.6	—	—	—	—	—	—
	0.85	0.226	167.0	64.7	—	—	—	—	—	—
	0.90	0.148	267.5	43.9	—	—	—	—	—	—

less bonus or slightly more penalty to achieve the same probability of on-time completion and a better expected tardiness.

2. When the task operator has midrange to large present bias ($\beta \in \{0.2, 0.4, 0.6\}$), the WMT scheme uses both less bonus and less penalty to achieve the target probability of on-time completion and a better expected tardiness. The smaller the value of β , the larger the improvement.

3. When the task operator has extremely large present bias ($\beta = 0$), the LC scheme fails.

Overall, for a single task, the WMT scheme is better than the LC scheme when present bias is midrange or large. The LC scheme is better than the WMT scheme when present bias is small or absent. A project company that fails to understand the prevalence of present bias in project workers may thus use the LC scheme with misplaced confidence, as is apparently the case in practice. Moreover, the small present bias of one task operator may only delay a single task slightly, which can be rectified by the LC scheme. However, in a project with many tasks, delay accumulates, making Cases II and III more likely to occur and the WMT scheme more effective. This is why the WMT scheme provides a still larger improvement than the LC scheme for both the serial network and the general network, as shown below.

5.3. Understanding the Strength of WMT

Our proposed incentive scheme, WMT, uses frequent monitoring of project progress S_i ; however, the widely used benchmark scheme LC is based on finishing time. A natural question is, does the relatively better performance of WMT result from frequent (and possibly costly or distracting) monitoring of project progress? That is, if we implement an LC bonus/penalty in each period (a scheme we denote by *period-LC*), does WMT still offer a significant improvement? We address this question by comparing the WMT and period-LC schemes in Appendix A. The main result is that the WMT scheme also performs much better than the period-LC scheme. The main strength of the WMT scheme lies in its efficient incentive function, which compensates for the present bias of the task operator, not in frequent monitoring. Moreover, both the classical LC and period-LC schemes use an incentive function that is inefficient, especially when the task operator suffers from severe present bias. Details appear in Section A.3 in Appendix A.

6. Serial Networks

Section 6.1 introduces a practical task time adjustment mechanism for serial networks. Section 6.2 presents a theoretical comparison between the performance of the WMT and LC schemes. Section 6.3 describes the

experimental design used in our computational studies here and in Section 7. Section 6.4 provides detailed computational results for random tasks.

6.1. Allowed Task Time Adjustment

During the completion of the serial network, we allow the project manager to adjust the allowed task time for the remaining tasks based on the performance of the previously finished tasks. When the previous tasks are on time, the project manager simply keeps the allowed task times of the remaining tasks unchanged. However, when the previous tasks are delayed, the project manager may attempt to “crash” the project by reducing the allowed task times of the remaining tasks in order to finish the whole project on time. Crashing is a standard approach to expediting delayed projects (Khang and Myint 1999, Klastorin 2012, Kerzner 2017). However, because of the scarcity of available human resources, legal restrictions on overtime, and physical or logistical lower bounds on task times, only limited crashing is available. For example, in the case study of Khang and Myint (1999), only one-third of the original planned task time can be reduced by increasing the working hours. Allowing for crashing cost and reduction in quality that we do not model, we assume that only one-fourth of the original planned task time can be reduced by crashing.³ More specifically, we adjust the allowed task time D_i for task i as follows:

$$D_i^{new} = \begin{cases} D_i - \min \left\{ \sum_{j=1}^{i-1} C_j - \sum_{j=1}^{i-1} D_j, \lfloor D_i/4 \rfloor \right\}, & \text{if } \sum_{j=1}^{i-1} C_j > \sum_{j=1}^{i-1} D_j, \\ D_i, & \text{if } \sum_{j=1}^{i-1} C_j \leq \sum_{j=1}^{i-1} D_j, \end{cases} \quad (9)$$

where C_j is the real finishing time of task j , which is the previous node of task i in the serial network, and $\lfloor D_i/4 \rfloor$ denotes the largest integer no bigger than $D_i/4$. As an example, suppose each task has expected duration 8, and the 10th task finishes at time 84, that is, $\sum_{j=1}^{10} C_j = 84$, instead of time $\sum_{j=1}^{10} D_j = 80$, as planned. Then, for the 11th task, the adjusted task time is $D_{11} = 8 - \min\{84 - 80, 8/4\} = 6$. The adjustment to allowed task times in (9) is implemented under both the WMT and LC schemes to ensure a fair comparison.

6.2. Theoretical Comparison

We provide a theoretical comparison of the WMT and LC schemes on a simple serial network with two tasks and then develop the result for the serial network with N tasks. For the WMT scheme, we introduce the following notation,

$$f_{WMT}^{i,D_i}(t) = Pr_{WMT}^{D_i}(\tau_i = t), \quad F_{WMT}^{i,D_i}(t) = Pr_{WMT}^{D_i}(\tau_i \leq t),$$

to denote the probability density function (PDF) and the cumulative density function (CDF) of the task time distribution of task i , under the WMT scheme with incentives specified by an allowed task time D_i , respectively. We also introduce corresponding notation $f_{LC}^{i,D_i}(t)$ and $F_{LC}^{i,D_i}(t)$ for the LC scheme. Figure 7 provides two typical CDFs under both the WMT and LC schemes at the task level, which are discussed in detail in Example 2. In Figure 7(a), for example, the WMT scheme (the solid line) performs better than the LC scheme (the dotted line) by generating a better task time distribution with larger CDF values. By specifying some conditions, we further prove that the good performance of the WMT scheme for the single task extends to the serial network.

For a serial network with two tasks and $s \geq D_1 + D_2$, we have

$$\begin{aligned} Pr_{WMT}(\tau_1 + \tau_2 \leq s) &= \sum_{k=1}^{D_1} f_{WMT}^{1,D_1}(k) F_{WMT}^{2,D_2}(s-k) \\ &\quad + \sum_{j=D_1+1}^{s-1} f_{WMT}^{1,D_1}(j) F_{WMT}^{2,D_2-1}(s-j), \\ Pr_{LC}(\tau_1 + \tau_2 \leq s) &= \sum_{k=1}^{D_1} f_{LC}^{1,D_1}(k) F_{LC}^{2,D_2}(s-k) \\ &\quad + \sum_{j=D_1+1}^{s-1} f_{LC}^{1,D_1}(j) F_{LC}^{2,D_2-1}(s-j). \end{aligned}$$

The following theorem provides sufficient conditions under which $Pr_{WMT}(\tau_1 + \tau_2 \leq s) \geq Pr_{LC}(\tau_1 + \tau_2 \leq s)$, for $s \geq D_1 + D_2$, which implies that the WMT scheme achieves larger probability of on-time completion and smaller expected tardiness than the LC scheme. In the theorem, we assume that $\lfloor D_i/4 \rfloor = 1$, that is, the allowed task time can only be adjusted by one period when the project is behind schedule.

Theorem 2. For a serial network with two tasks, if the WMT scheme achieves better performance for CDF than the LC scheme as long as t is no smaller than the allowed time for the second task, that is,

$$F_{WMT}^{2,D_2}(t) \geq F_{LC}^{2,D_2}(t), \quad t = D_2, D_2 + 1, \dots, T; \quad (10)$$

and meanwhile the following condition holds, for $s \geq D_1 + D_2$,

$$\begin{aligned} G(s) &= \sum_{k=1}^{D_1} [f_{WMT}^{1,D_1}(k) - f_{LC}^{1,D_1}(k)] F_{LC}^{2,D_2}(s-k) \\ &\quad + \sum_{j=D_1+1}^{s-1} [f_{WMT}^{1,D_1}(j) F_{WMT}^{2,D_2-1}(s-j) - f_{LC}^{1,D_1}(j) F_{LC}^{2,D_2-1}(s-j)] \geq 0, \end{aligned} \quad (11)$$

then, for the network, we have

$$Pr_{WMT}(\tau_1 + \tau_2 \leq s) \geq Pr_{LC}(\tau_1 + \tau_2 \leq s), \quad s \geq D_1 + D_2,$$

which implies that the WMT scheme has a larger probability of on-time completion and a smaller expected tardiness than the LC scheme.

The conditions introduced in Theorem 2 are in fact mild. The following example shows that these conditions are easily satisfied when $\beta = 0.4$ or $\beta = 0.6$.

Example 2. We consider a serial network with two identical tasks. The parameter settings follow those in Example 1. First, we choose $\beta = 0.4$, $b = p = 12.5$, and $B = P = 175$. From the first line in panel A of Table 3, the WMT scheme achieves larger probability of on-time completion and smaller expected tardiness than the LC scheme, with less bonus and penalty for the single task. The CDFs of the finishing time under the WMT and LC schemes are further shown in Figure 7(a), which reveals that the finishing time under the WMT scheme is better in the sense of first-order stochastic dominance and the WMT scheme works more efficiently than the LC scheme. The reason lies in the setting $\beta = 0.4$, describing

Figure 7. (Color online) CDF of the Task Time for a Single Task Under the WMT and LC Schemes

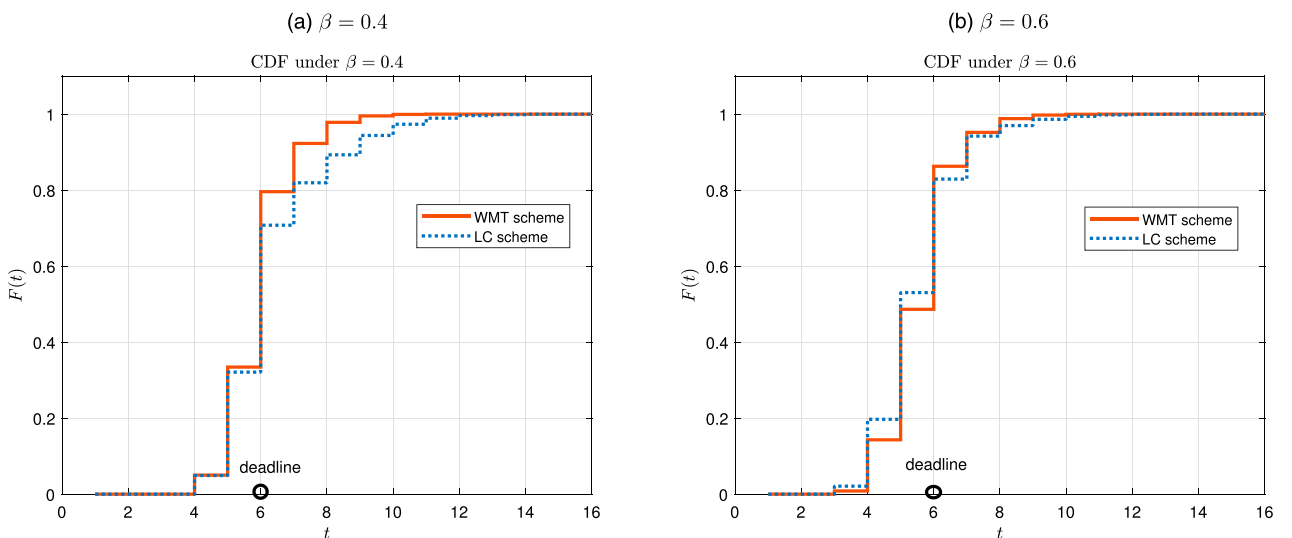


Table 3. Performance Comparison Between WMT and LC Schemes for a Serial Network with Two Identical Tasks

Panel A: Single task performance and two tasks performance when $\beta = 0.4$									
Project type	WMT				LC				
	$Pr(\tau \leq D)$	Tardiness	Bonus	Penalty	$Pr(\tau \leq D)$	Tardiness	Bonus	Penalty	
Single task $D = 6$	0.7960	0.0919	64.4923	35.4771	0.7077	0.2237	65.7658	118.7900	
Two tasks	0.7828	0.3526	127.8014	77.0453	0.6421	0.9098	128.9120	238.8435	
Panel B: Condition check when $\beta = 0.4$									
k	$Pr_{WMT}^6(\tau \leq k)$	$Pr_{LC}^6(\tau \leq k)$	Diff	s	$G(s)$	t	$Pr_{WMT}(\tau_P \leq t)$	$Pr_{LC}(\tau_P \leq t)$	Diff
3	0	0	0	12	0.0729	7	0	0	0
4	0.0501	0.0490	0.0010	13	0.0692	8	0.0025	0.0021	0.0004
5	0.3344	0.3211	0.0133	14	0.0638	9	0.0300	0.0294	0.0006
6	0.7960	0.7077	0.0883	15	0.0642	10	0.1629	0.1484	0.0145
7	0.9229	0.8194	0.1035	16	0.0685	11	0.4613	0.3975	0.0639
8	0.9784	0.8930	0.0855	17	0.0746	12	0.7828	0.6421	0.1407
9	0.9953	0.9439	0.0514	18	0.0797	13	0.9093	0.7641	0.1452
10	0.9993	0.9734	0.0259	19	0.0834	14	0.9676	0.8510	0.1166
11	1	0.9894	0.0105	20	0.0856	15	0.9906	0.9134	0.0772
12	1	0.9962	0.0037	21	0.0869	16	0.9977	0.9520	0.0456
13	1	0.9987	0.0013	22	0.0877	17	0.9993	0.9753	0.0240
14	1	0.9997	0.0003	23	0.0880	18	0.9999	0.9876	0.0123
15	1	0.9999	0.0001	24	0.0882	19	1	0.9938	0.0062
Panel C: Single task performance and two tasks performance when $\beta = 0.6$									
Project type	WMT				LC				
	$Pr(\tau \leq D)$	Tardiness	Bonus	Penalty	$Pr(\tau \leq D)$	Tardiness	Bonus	Penalty	
Single task $D = 6$	0.7962	0.0922	50.9471	25.6628	0.7775	0.0885	53.1069	38.5287	
Two tasks	0.7818	0.3555	100.6082	57.1233	0.7818	0.4124	102.3204	80.6075	
Panel D: Condition check when $\beta = 0.6$									
k	$Pr_{WMT}^6(\tau \leq k)$	$Pr_{LC}^6(\tau \leq k)$	Diff	s	$G(s)$	t	$Pr_{WMT}(\tau_P \leq t)$	$Pr_{LC}(\tau_P \leq t)$	Diff
3	0	0	0	12	0.0075	7	0	0	0
4	0.0420	0.0923	-0.0503	13	0.0174	8	0.0018	0.0080	-0.0062
5	0.3450	0.4135	-0.0685	14	0.0182	9	0.0271	0.0646	-0.0374
6	0.7962	0.7775	0.0187	15	0.0151	10	0.1584	0.2416	-0.0831
7	0.9202	0.9197	0.0004	16	0.0148	11	0.4672	0.5312	-0.0640
8	0.9775	0.9615	0.0160	17	0.0159	12	0.7818	0.7818	0.0000
9	0.9952	0.9836	0.0116	18	0.0171	13	0.9080	0.8987	0.0093
10	0.9991	0.9937	0.0055	19	0.0179	14	0.9681	0.9483	0.0199
11	1	0.9979	0.0020	20	0.0184	15	0.9900	0.9758	0.0143
12	1	0.9994	0.0005	21	0.0186	16	0.9972	0.9897	0.0075
13	1	0.9999	0.0001	22	0.0187	17	0.9994	0.9959	0.0035
14	1	1	0	23	0.0187	18	0.9999	0.9985	0.0014

Notes. For $\beta = 0.4$, we choose $b = p = 12.5$ and $B = P = 175$; for $\beta = 0.6$, we choose $b = p = 9.5$ and $B = P = 105$. Diff, difference.

severe present bias. When $\beta = 0.4$, the Case I, Case II, and Case III conditions in Theorem 1 are all satisfied; that is, the WMT scheme motivates a larger effort level than the LC scheme for all three cases; hence, the task time under the WMT scheme is typically smaller than under the LC scheme.⁴

The column “Diff” in panel B of Table 3 shows the difference between the previous two columns. The first six columns in panel B of Table 3 confirm that the parameter setting satisfies the conditions given in Theorem 2. Thus, based on Theorem 2, we conclude that the WMT scheme achieves higher probability of on-time

completion and smaller expected tardiness than the LC scheme with less bonus and penalty for the network of two tasks, which is also confirmed by the second line in panel A of Table 3. In fact, as shown in the last three columns in panel B of Table 3, the finishing time distribution of the network under the WMT scheme is also better in the sense of first-order stochastic dominance.

Second, we choose $\beta = 0.6$, $b = p = 9.5$ and $B = P = 105$. From the first line in panel C of Table 3, the WMT scheme achieves larger probability of on-time completion and smaller expected tardiness than the LC scheme with less bonus and penalty for the single

task. The CDFs of the task time under both WMT and LC are also shown in Figure 7(b). When $\beta = 0.6$, the WMT scheme motivates a larger effort level than the LC scheme in Cases II and III.⁵ This implies that the task time under the WMT scheme may not be smaller in the sense of first-order stochastic dominance. But the WMT scheme typically has better performance than the LC scheme at and after the deadline. Figure 7(b) confirms this intuition.

The first six columns in panel D of Table 3 reveal that the parameter setting satisfies the conditions given in Theorem 2. Thus, based on Theorem 2, we conclude that $Pr_{WMT}(\tau_1 + \tau_2 \leq s) \geq Pr_{LC}(\tau_1 + \tau_2 \leq s)$ for $s \geq D_1 + D_2$, which is confirmed by the last three columns in panel D of Table 3. Furthermore, the WMT scheme achieves higher probability of on-time completion and smaller expected tardiness than the LC scheme with less bonus and penalty for the serial network of two tasks, as confirmed by the second line in panel C of Table 3.

Comparing the case $\beta = 0.6$ with $\beta = 0.4$, we find that the total spending (penalty and bonus) is higher for $\beta = 0.4$ under both WMT and LC. The intuition here is straightforward: in order to achieve the same target probability of on-time completion, motivating a task operator with severe present bias is more expensive.

Next, we provide sufficient conditions under which the WMT scheme has better performance than the LC scheme for a serial network with N tasks.

Theorem 3. *If the WMT scheme achieves better performance for CDF than the LC scheme for each task in the serial network, that is,*

$$\begin{aligned} F_{WMT}^{1,D_1}(t) &\geq F_{LC}^{1,D_1}(t), \quad t = 1, 2, \dots, T, \\ F_{WMT}^{i,D}(t) &\geq F_{LC}^{i,D}(t), \quad t = 1, 2, \dots, T, \quad i = 2, 3, \dots, N, \\ D &= D_i, D_i - 1, \end{aligned}$$

and meanwhile the LC scheme satisfies

$$F_{LC}^{i,D_i}(t) \geq F_{LC}^{i,D_i-1}(t-1), \quad t = 1, 2, \dots, T, \quad i = 2, 3, \dots, N,$$

then for the serial network of N tasks, we have

$$Pr_{WMT}\left(\sum_{i=1}^N \tau_i \leq t\right) \geq Pr_{LC}\left(\sum_{i=1}^N \tau_i \leq t\right), \quad t = 1, 2, \dots, NT,$$

which implies that the WMT scheme has a larger probability of on-time completion and a smaller expected tardiness than the LC scheme.

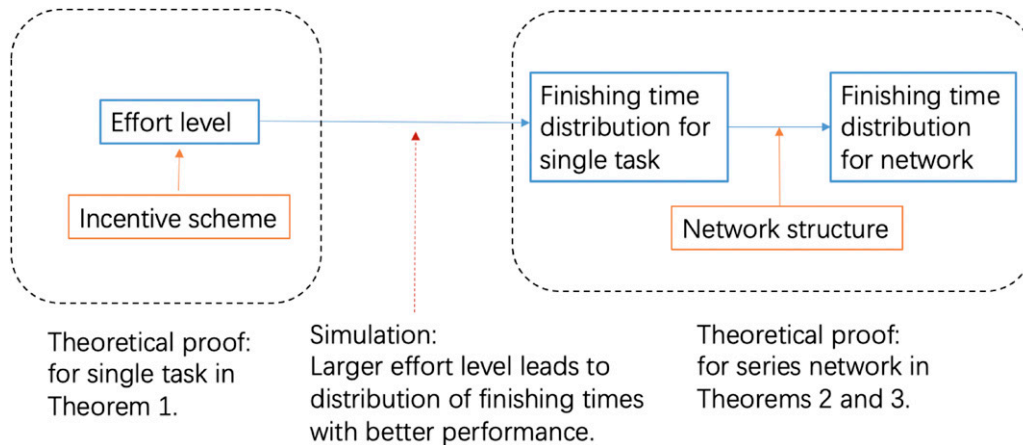
Remark 1. The condition in Theorem 3 requires that the individual task time under the LC scheme first order stochastically dominates that under WMT. This condition is stronger than the condition in Theorem 2.

Remark 2. Figure 8 illustrates the logical flow that underlies Theorems 1, 2 and 3. Theorem 1 provides necessary and sufficient conditions under which $e^{WMT} \geq e^{LC}$. Theorems 2 and 3 provide sufficient conditions under which the relative performance of WMT and LC remains the same from a single task to a simple serial network with two tasks and N tasks, respectively. The simulation results shown in Figure 7 and Table 2 complete the link between Theorem 1 and Theorems 2 and 3 by showing that a larger effort level generally leads to better finishing time distribution.

6.3. Experimental Design

We describe the experimental design of our main computational study of project performance for both serial (as in Section 6.4) and general networks (as in Section 7.1). We need to define the following: (i) the distribution of random progress variables Q_i ; (ii) the distribution of empirical β values among the task operators; (iii) the random assignment of operators to tasks; and (iv) the randomly generated network. First, Q_t models

Figure 8. (Color online) Flowchart of Theoretical Proofs



practical uncertainty in the mapping of effort level e_t into project progress. Similar to Example 1 in Section 5.2, we assume Q_t to be uniformly distributed, $q_i = i/4$ ($i = 1, \dots, 7$), $p_i = 1/7$ ($i = 1, \dots, 7$).

Second, the value(s) of β , as shown in Table 2, play an important role in the relative performance of the WMT and LC schemes. Based on the empirical results summarized in Table 1, we let β be normally distributed across the population, $N(0.65, 0.08^2)$. For ease of computation, we approximate this normal distribution with the discrete distribution:

$$\begin{aligned} Pr(\beta_i = 0.65 - 0.08 \times 3) &= N(0.5 - 3), \\ Pr(\beta_i = 0.65 + 0.08 \times j) &= N(0.5 + j) - N(-0.5 + j), \\ &\quad j = -2, \dots, 2, \\ Pr(\beta_i = 0.65 + 0.08 \times 3) &= 1 - N(-0.5 + 3), \end{aligned} \quad (12)$$

where $N(\cdot)$ is the cumulative probability distribution function of a standard normal random variable. In doing so, we are sampling from $[\mu - 3\sigma, \mu + 3\sigma]$ with seven points in total.

Third, given a set of operators with various values of β , the assignment of operators to tasks significantly affects project performance. For example, assigning the task operators with higher β values to more complex tasks typically improves project performance. We name this property the *assignment effect*. The task assignment problem is not the focus of this work, but we discuss it as a direction for future research in Section 8. However, in many situations the project manager does not know the β value of an individual task operator. Hence, we randomly assign the tasks to the operators and then repeat the random assignment procedure 100 times to average out the assignment effect.

Finally, we randomly generate projects with different network structures 100 times, from serial (as in Section 6.4) to general (as in Section 7) networks. This controls for the possibility that performance is affected by random network structure.

Both here and in Section 7, we use the following settings for other features. The workload of the task i , S_i , is $8D_i$. The maximum effort level in each period is 12, $e^u = 12$. We set $\gamma = 10$, $\delta = 0.995$, $u_1(x) = x^2$, $u_2(x) = x^2$. The marginal bonus level and the marginal penalty level under the WMT scheme are the same, that is, $b = p$. Also, the marginal bonus level and the marginal penalty level under the LC scheme are the same, that is, $B = P$. Each task in the network can have seven different expected durations: 6, 7, ..., 12, with equal probability one-seventh.

6.4. Computational Comparison

We set the number of tasks (N) in a serial network as $N \in \{50, 100, 150, 200\}$. We randomly generate 100 serial networks for each choice of number of tasks.

Further, for each network we randomly assign the tasks to task operators whose β values are also randomly generated by the discrete distribution in (12). We then repeat the task assignment procedure 100 times to control for any assignment effect. Therefore, we have 100×100 different realizations for each network structure in total.

Given the network structure and task assignment, we compute the accumulated bonus (*bonus*) and the accumulated penalty (*penalty*) in order to achieve the target probability of on-time completion. We also record the corresponding expected tardiness (*tardiness*). Then, we compute the average values of the accumulated bonus, the accumulated penalty, and the expected tardiness over 100×100 realizations. We expect that these average results represent the robust performance of the WMT and LC schemes in practice. Table 4 reports the performance of the WMT and LC schemes with this design. Compared with the LC scheme, the WMT scheme consistently uses less bonus and less penalty to achieve the same target probability of on-time completion and smaller expected tardiness for projects across our experimental design.

7. General Networks

Section 7.1 applies the WMT and LC incentive schemes to a variety of randomly generated general acyclic project networks. Section 7.2 studies the sensitivity of performance to various project network characteristics.

7.1. Computational Comparison

We generate networks using the random network generation software RanGen (Demeulemeester et al. 2003), available at <http://www.projectmanagement.ugent.be/research/data/RanGen>. We specify the number of tasks (N) and the order strength (OS), which is the number of direct precedence relationships in the network divided by the theoretical maximum number.

Because of the more complicated structure of general networks compared with serial ones, the adjustment mechanism for allowed task times under crashing, analogous to (9), requires a different form. Thus, the adjusted allowed time of task j is

$$D_j^{new} = \begin{cases} D_j - \min\{T_R^j - T_E^j, \lfloor D_j/4 \rfloor\}, & \text{if } T_R^j > T_E^j, \\ D_j, & \text{if } T_R^j \leq T_E^j, \end{cases} \quad (13)$$

where T_R^j is the realized start time of task j and T_E^j is the expected start time of task j . In general, for the tasks that are not on the critical path, we choose the expected late start time as the expected start time. When the remaining time is less than D_j , the project manager reduces the allowed time of task j . The

Table 4. Performance Comparison Between WMT and LC Schemes for Serial Networks with Random Tasks and Random Assignment

N	$Pr(\tau \leq D)$	WMT			LC			Improvement		
		Tardiness	Bonus	Penalty	Tardiness	Bonus	Penalty	Tardiness	Bonus	Penalty
50	0.80	0.325	554.8	1,304.7	0.352	591.5	1,305.4	7.5%	6.2%	0.0%
	0.85	0.242	766.0	1,368.1	0.264	804.7	1,424.4	8.4%	4.8%	4.0%
	0.90	0.159	1,043.6	1,408.0	0.176	1,094.1	1,527.8	9.2%	4.6%	7.8%
	0.95	0.079	1,441.7	1,402.8	0.087	1,554.7	1,632.9	9.6%	7.3%	14.1%
	0.99	0.016	2,273.8	1,341.6	0.017	2,450.1	1,607.6	10.7%	7.2%	16.5%
100	0.80	0.325	1,128.5	2,653.2	0.360	1,273.3	2,675.7	9.6%	11.4%	0.8%
	0.85	0.242	1,522.3	2,756.7	0.270	1,666.3	2,876.0	10.4%	8.6%	4.2%
	0.90	0.159	2,007.6	2,797.5	0.179	2,158.9	3,043.9	11.1%	7.0%	8.1%
	0.95	0.079	2,622.5	2,778.5	0.089	2,830.9	3,215.9	11.6%	7.4%	13.6%
	0.99	0.016	3,839.9	2,705.8	0.018	4,042.0	3,142.6	12.2%	5.0%	13.9%
150	0.80	0.327	1,692.3	3,985.5	0.358	1,912.7	4,021.6	8.8%	11.5%	0.9%
	0.85	0.243	2,272.6	4,129.0	0.269	2,477.3	4,308.6	9.7%	8.3%	4.2%
	0.90	0.160	2,967.9	4,183.5	0.179	3,160.2	4,543.7	10.5%	6.1%	7.9%
	0.95	0.079	3,785.1	4,153.0	0.089	4,130.5	4,787.0	11.1%	8.4%	13.2%
	0.99	0.016	5,331.1	4,082.6	0.018	5,652.1	4,656.5	11.4%	5.7%	12.3%
200	0.80	0.326	2,266.8	5,321.1	0.356	2,568.3	5,379.7	8.4%	11.7%	1.1%
	0.85	0.243	3,035.2	5,502.5	0.267	3,308.7	5,750.6	9.3%	8.3%	4.3%
	0.90	0.160	3,937.5	5,569.7	0.178	4,188.1	6,050.7	10.0%	6.0%	7.9%
	0.95	0.080	4,936.8	5,527.3	0.089	5,423.0	6,362.3	10.6%	9.0%	13.1%
	0.99	0.016	6,767.2	5,465.5	0.018	7,253.8	6,168.4	11.6%	6.7%	11.4%

adjustment to allowed task time in (13) is implemented under both the WMT and LC schemes.

In the computational study, we follow the same assumptions about Q_i , β , and random assignment of operators to the tasks as made in Section 6.3. We set $N \in \{100, 200\}$, $OS \in \{0.25, 0.50, 0.75\}$. For each combination of N and OS , we randomly generate 100 different networks and randomly assign different β values to the operators. Table 5 reports the accumulated bonus and accumulated penalty required to achieve a given target probability of on-time completion as well as expected tardiness values.

From the results in Table 5, we observe the following: (i) The WMT scheme has robustly better performance than the LC scheme across different general networks. Meanwhile, the better performance of WMT comes at lower cost as measured by smaller bonus and penalty. (ii) The WMT scheme has a larger advantage than the LC scheme when N is large. That is, the WMT scheme performs better for more complex networks.

7.2. Sensitivity Analysis

We conduct a sensitivity analysis to study the relative performance of the WMT and LC schemes under various network structures. We vary the number of tasks and the order strength to see how network structure affects the performance of the two schemes. We let $N \in \{20, 40, \dots, 200\}$ and $OS \in \{0.1, 0.2, \dots, 1\}$, which generates 100 different network types. For each network type, we randomly generate 100 networks. Finally, for each randomly generated network, we repeat the task assignment procedure 100 times to control for assignment effect. We set

the target probability of on-time completion equal to 0.95. We compute the average accumulated bonus, average accumulated penalty, and average expected tardiness to achieve this target probability.

We first study the effect of the number of tasks on the three performance measures, as shown in Figure 9. From Figure 9, the accumulated bonus and accumulated penalty under both the WMT and LC schemes are all increasing functions of N . The intuition here is straightforward: in order to achieve the target probability of on-time completion, the larger the network, the larger the cost in terms of both penalty and bonus. However, the increase is slower under WMT than under LC, which makes the absolute improvements increasing functions of N . Thus, the larger the project network, the better the performance of the WMT scheme. Meanwhile, the WMT scheme achieves the same probability of on-time completion, but with smaller expected tardiness and at smaller cost, than the LC scheme.

The order strength of a network is the number of precedence relations, including transitive ones but excluding arcs incident to dummy start and end tasks, divided by the theoretical maximum number of precedence relations, which is $n(n-1)/2$. Figure 10 shows the effect of OS on the three performance measures. From Figure 10, we observe that the expected tardiness under the LC scheme is a constant function of OS ; the accumulated bonus under the LC scheme is a decreasing function of OS ; and the accumulated penalty under the LC scheme is an increasing function of OS . Meanwhile, the accumulated bonus under the

Table 5. Performance Comparison Between WMT and LC Schemes for General Networks

Case	$Pr(\tau \leq D)$	WMT			LC			Improvement		
		Tardiness	Bonus	Penalty	Tardiness	Bonus	Penalty	Tardiness	Bonus	Penalty
$N = 100$ $OS = 0.25$	0.80	0.313	2,719	2,628	0.353	4,621	3,126	11.2%	41.2%	15.9%
	0.85	0.232	3,647	2,638	0.263	6,367	3,208	11.9%	42.7%	17.8%
	0.90	0.152	5,066	2,554	0.175	9,010	3,198	12.8%	43.8%	20.1%
	0.95	0.074	7,708	2,410	0.088	14,017	3,074	15.2%	45.0%	21.6%
	0.99	0.015	13,170	2,243	0.018	28,171	2,597	15.7%	53.3%	13.6%
$N = 100$ $OS = 0.5$	0.80	0.317	2,553	2,647	0.360	4,067	3,097	12.0%	37.2%	14.5%
	0.85	0.235	3,296	2,656	0.268	5,371	3,189	12.5%	38.6%	16.7%
	0.90	0.154	4,390	2,600	0.178	7,341	3,216	13.3%	40.2%	19.2%
	0.95	0.075	6,329	2,490	0.088	10,870	3,180	14.9%	41.8%	21.7%
	0.99	0.015	11,046	2,287	0.018	20,313	2,796	17.0%	45.6%	18.2%
$N = 100$ $OS = 0.75$	0.80	0.322	1,972	2,675	0.363	2,737	2,919	11.3%	28.0%	8.3%
	0.85	0.238	2,578	2,694	0.270	3,598	3,066	11.9%	28.3%	12.1%
	0.90	0.157	3,402	2,694	0.179	4,858	3,180	12.5%	30.0%	15.3%
	0.95	0.077	4,811	2,588	0.089	7,149	3,225	13.6%	32.7%	19.8%
	0.99	0.015	8,058	2,387	0.018	12,709	3,140	16.1%	36.6%	24.0%
$N = 200$ $OS = 0.25$	0.80	0.313	6,982	5,014	0.354	12,135	6,272	11.4%	42.5%	20.1%
	0.85	0.230	8,817	5,074	0.263	15,600	6,354	12.6%	43.5%	20.1%
	0.90	0.151	11,489	5,067	0.174	20,686	6,307	13.1%	44.5%	19.7%
	0.95	0.074	16,370	4,862	0.087	30,037	6,071	14.8%	45.5%	19.9%
	0.99	0.015	29,720	4,433	0.017	53,293	5,112	15.7%	44.2%	13.3%
$N = 200$ $OS = 0.5$	0.80	0.316	5,165	5,145	0.358	9,040	6,216	11.5%	42.9%	17.2%
	0.85	0.233	6,503	5,185	0.266	11,481	6,363	12.2%	43.4%	18.5%
	0.90	0.153	8,365	5,176	0.176	15,152	6,442	13.1%	44.8%	19.7%
	0.95	0.075	11,669	5,034	0.087	21,590	6,356	13.9%	46.0%	20.8%
	0.99	0.015	20,105	4,613	0.018	38,669	5,706	17.3%	48.0%	19.1%
$N = 200$ $OS = 0.75$	0.80	0.322	3,603	5,229	0.364	5,681	5,859	11.5%	36.6%	10.8%
	0.85	0.238	4,564	5,303	0.271	7,190	6,116	12.2%	36.5%	13.3%
	0.90	0.156	5,847	5,318	0.179	9,379	6,328	12.7%	37.7%	16.0%
	0.95	0.077	7,986	5,270	0.089	13,319	6,470	13.5%	40.0%	18.6%
	0.99	0.015	12,936	4,965	0.018	22,569	6,375	15.6%	42.7%	22.1%

WMT scheme is also a decreasing function of OS but with smaller negative slope. The expected tardiness and accumulated penalty under the WMT scheme are increasing functions of OS . Thus, the absolute improvements of the WMT scheme relative to the LC scheme in expected tardiness and accumulated bonus are decreasing functions of OS . However, the absolute improvements of the WMT scheme relative to the LC scheme for accumulated penalty are initially increasing and then decreasing function of OS and overall a nearly constant function of OS .

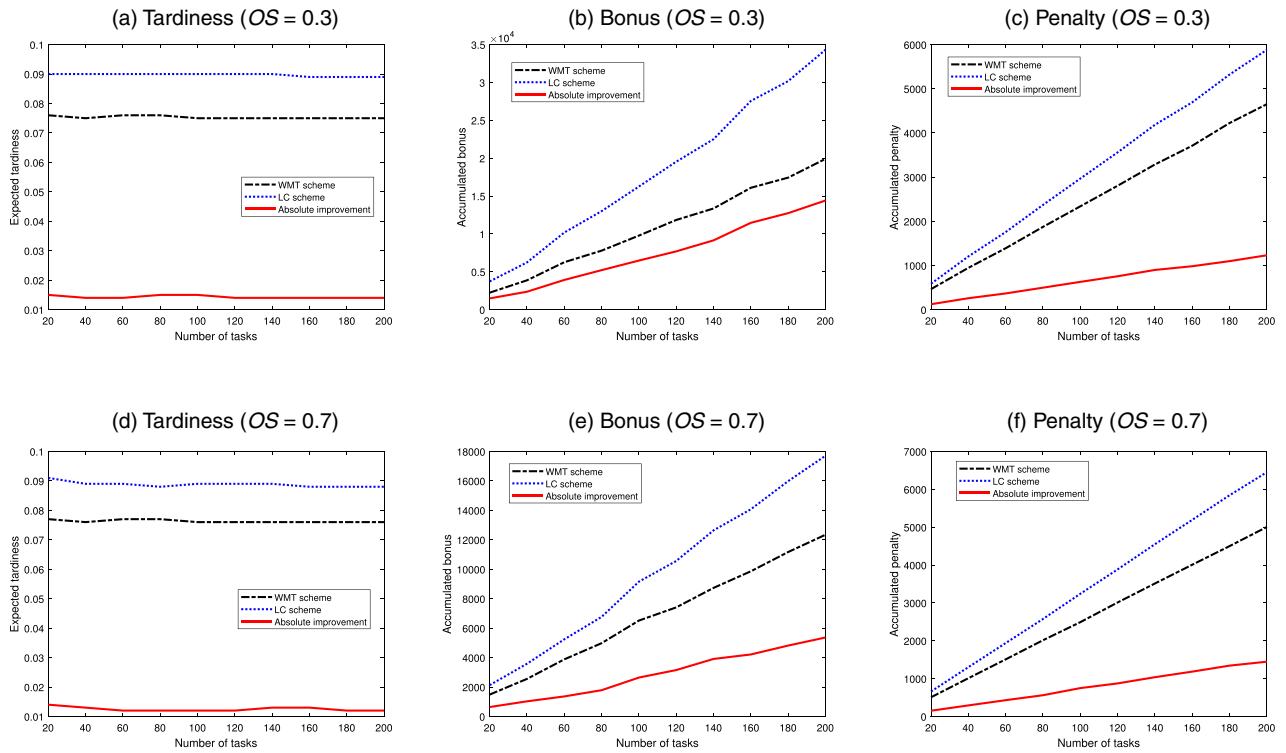
We provide further insights about these results. Figure 11 shows the relative improvements of the WMT scheme for the three performance measures with respect to N and OS . Summarizing, when either N increases or OS decreases, the relative improvement of the WMT scheme in accumulated bonus increases. The relative improvement of the WMT scheme in expected tardiness is a decreasing function of OS . The relative improvement of the WMT scheme in accumulated penalty is a constant function of N and achieves its maximum value around $OS = 0.5$. Although Figure 11(c) achieves its maximum around $OS = 0.5$, the graph is almost constant in the range $0 \leq OS \leq 0.5$ but decreases quickly when $0.5 \leq OS \leq 1$. This is consistent

with the negative coefficient (-0.054) of relative improvement in penalty as OS varies in Table 6.

Finally, we provide a regression analysis for the above results, as shown in Table 6. The numbers in parentheses are p -values. The results in Table 6 fully confirm those in Figures 9–11. For both the relative and absolute improvement, the coefficients of OS are significantly negative. Thus, the WMT scheme becomes relatively stronger as OS decreases. For the bonus and penalty measures, the coefficients of N on both the relative and absolute improvements are positive, thus the relative strength of WMT becomes larger as N increases. For the absolute improvement of the WMT scheme in expected tardiness, the coefficient of N is significantly negative but is very small (-0.001). This observation echoes the almost constant graphs in Figure 9(a), which show that tardiness does not vary much with N .

8. Conclusions

This work addresses the globally observed and financially damaging problem of late delivery of projects. Our review of the empirical evidence, and the business and academic literatures, suggests that existing incentive schemes fail to eliminate the problem of late

Figure 9. (Color online) Performance Comparison of WMT and LC Schemes with Various Numbers of Tasks

project execution. Our explanation for this, based on our analysis of empirical data, is that those schemes fail to address workers' time-inconsistent preferences,

or present bias. Our work is among the first to model present bias across multiple periods in the context of a general project network.

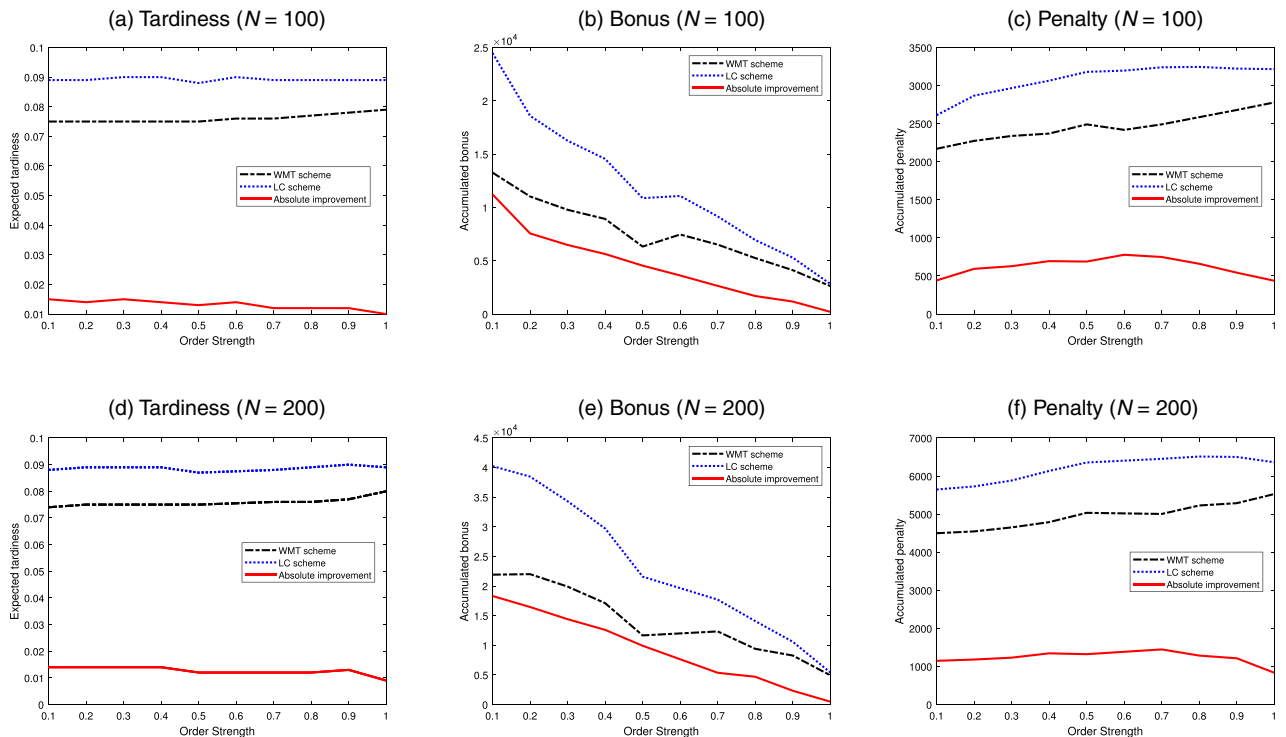
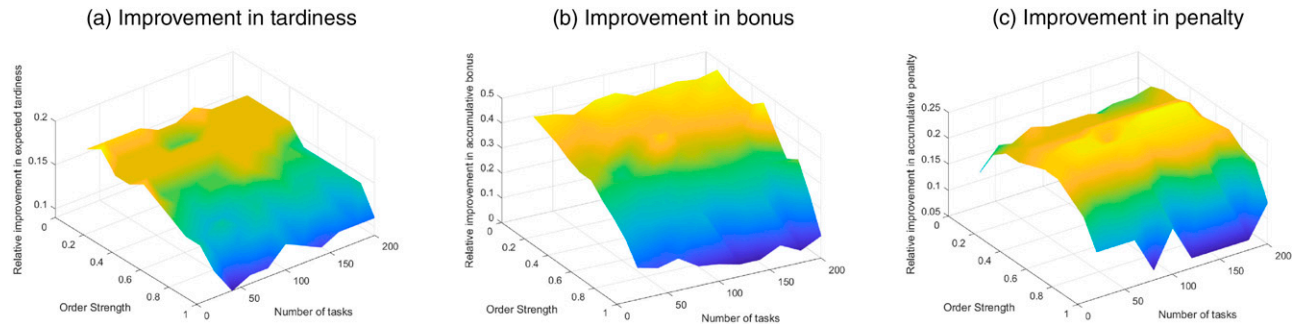
Figure 10. (Color online) Performances Comparison of WMT and LC Schemes for Various Order Strengths

Figure 11. (Color online) Relative Improvements of WMT Scheme in Expected Tardiness, Accumulated Bonus, and Accumulated Penalty



We design an incentive scheme, Work More Tomorrow, to mitigate present bias in project execution. This scheme can easily be implemented on a spreadsheet. The performance of our scheme is compared with that of a traditional linear contract for three environments: single tasks, serial networks, and general acyclic project networks. In the first two environments, we provide a theoretical comparison between our incentive scheme and the benchmark. In all three environments, we show computationally that our incentive scheme provides a significant advantage that is robust across various project characteristics. A statistical analysis

shows that the relative advantage of our scheme typically increases with project size.

The development and analysis of our incentive scheme enable us to identify several insights that should be of value to project managers. First, our sensitivity analysis results enable project managers to understand the characteristics of projects for which the relative advantage of our incentive scheme is most valuable. Second, in a situation of present bias, frequent evaluation of project progress and related reward become more valuable and worthwhile. Third, managers should try to understand the present bias of

Table 6. Regression Results

Dependent variable	Coefficients of independent variables			Adjusted R^2	F-statistics
	Constant	OS	$N/200$		
<i>Tardiness under WMT scheme</i>	0.074*** (6.21E-145)	0.004*** (1.92E-30)	−0.001*** (0.007)	0.771	156.00 (5.63E-30)
<i>Bonus under WMT scheme</i>	6,502.5*** (4.21E-22)	−10,970.8*** (2.00E-31)	14,190.5*** (2.30E-39)	0.905	437.37 (4.39E-47)
<i>Penalty under WMT scheme</i>	−343.5*** (2.64E-19)	612.2*** (1.26E-29)	4,926.8*** (1.55E-105)	0.995	9,119.85 (1.26E-104)
<i>Tardiness under LC scheme</i>	0.091*** (2.21E-146)	−0.001*** (7.02E-05)	−0.001*** (1.09E-05)	0.281	18.94 (1.37E-07)
<i>Bonus under LC scheme</i>	12,447.0*** (4.29E-21)	−2,2054.7*** (1.07E-31)	23,466.6*** (2.59E-33)	0.885	356.08 (1.78E-43)
<i>Penalty under LC scheme</i>	−280.3*** (6.38E-08)	499.5*** (1.65E-13)	6,145.4*** (3.67E-96)	0.992	5,593.58 (3.91E-95)
<i>Absolute improvement in tardiness</i>	0.016*** (4.81E-75)	−0.005*** (7.55E-29)	−0.001*** (0.009)	0.748	137.42 (4.43E-28)
<i>Absolute improvement in bonus</i>	5,944.5*** (5.75E-19)	−11,083.9*** (1.53E-30)	9,276.1*** (4.08E-25)	0.851	263.75 (2.30E-38)
<i>Absolute improvement in penalty</i>	63.1 (0.228)	−112.7* (0.077)	1,218.6*** (1.79E-33)	0.801	185.98 (1.08E-32)
<i>Relative improvement in tardiness</i>	0.178*** (4.76E-78)	−0.056*** (6.97E-30)	−0.003 (0.359)	0.758	144.95 (7.17E-29)
<i>Relative improvement in bonus</i>	0.496*** (7.74E-60)	−0.351*** (1.06E-40)	0.027* (0.077)	0.862	288.13 (7.54E-40)
<i>Relative improvement in penalty</i>	0.229*** (4.57E-35)	−0.054*** (1.92E-04)	−0.002 (0.906)	0.125	7.57 (9.18E-04)

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

their task operators, and also the operators' level of awareness about it, in order to make more effective task assignments. Intuitively, task operators with larger present bias, especially if they are naive about it, should not be assigned to a longer or more complex task. Fourth, because most companies run multiple projects concurrently, similar insights can be applied to assignment of operators among projects.

We now discuss the implementability of our incentive scheme, WMT. Fotsch and Case (2015) describe and compare various types of incentive plans. They identify four characteristics of a successful incentive scheme. It should be (a) based on objective performance, (b) wholly transparent, (c) generous, and (d) self-funding in that the incentives paid are at least compensated on the revenue side. We note that our WMT scheme is easily implementable to provide all four of these characteristics. Fotsch and Case (2015) describe a successful incentive scheme that is implemented in a Los Angeles area company that manages small projects using weekly evaluations, corresponding to a period of one week in our incentive scheme.

Several related issues remain open for future research. First, it would be valuable to study the performance of our incentive scheme for specific project management applications. Second, our computational study shows that the assignment of operators to the tasks can influence performance considerably. Therefore, it would be valuable to integrate our incentive scheme with an effective algorithm for assigning operators to tasks based on their identified present bias. Third, following Wu et al. (2014), the effect of combining individuals into teams, and the resulting present bias in team decision making, can be studied using our incentive scheme. Finally, the extensions for naive decision makers considered by O'Donoghue and Rabin (2001), including partial naivete and a menu of choices, can be incorporated into a more general study. Similar to naive task operators, partially naive task operators also hold wrong beliefs about their future preferences. Therefore, the WMT scheme would still be effective in improving the performance of partially naive task operators. In conclusion, we hope that our work will encourage interest in this topic that has such an important impact on the world economy.

Appendix A. Comparison of the WMT and Period-LC Schemes

This section investigates how much of the improvement in performance comes from more frequent monitoring and how much comes from the WMT incentive scheme itself. That is, if we implement an LC bonus/penalty in each period (a scheme we denote by period-LC (PLC)), does WMT still offer a significant improvement? We address this question by comparing the WMT and period-LC schemes at the task level.⁶

A.1. Period Linear Contract Scheme

In order to implement an LC bonus/penalty in each period, we assume the manager can break down all tasks into subtasks and set deadlines for them. For simplicity, we consider the period-LC scheme for a single task. For a given task with workload S_1 and deadline D days, the manager divides the task into D subtasks with workload $\frac{S_1}{D}$ and deadline of one day. Thus, the task progress target at the end of day i can be set as⁷

$$S_{i+1}^{tar-PLC} = \frac{S_1(D-i)}{D}, \quad i = 1, 2, \dots, D.$$

Given the progress target, we denote the finishing time of subtask i as follows:

$$\tau_i = \min\{t : S_{t+1} \leq S_{i+1}^{tar-PLC}\}, \quad i = 1, 2, \dots, D.$$

The bonus for early completion and the penalty for late completion are defined for all D subtasks as follows:

$$\sum_{i=1}^D (B1_{\{\tau_i \leq i\}} + P1_{\{\tau_i > i\}})(i - \tau_i),$$

where $B > 0$ and $P > 0$ are a bonus and penalty per unit time, respectively. Also, the bonus and penalty occur at the finish times of each subtask. We denote this scheme as the period linear contract scheme.

Observing that $\{\tau_i = t\}$ is the same as $\{S_t > S_{i+1}^{tar-PLC}, S_{t+1} \leq S_{i+1}^{tar-PLC}\}$, we denote the bonus and penalty received at period t as follows,

$$I_t^{PLC} = \sum_{i=1}^D (B1_{\{t \leq i\}} + P1_{\{t > i\}})(i - t)1_{\{S_t > S_{i+1}^{tar-PLC}\}}1_{\{S_{t+1} \leq S_{i+1}^{tar-PLC}\}},$$

and the cumulative incentive through period t is

$$NI_{t+1}^{PLC} = \sum_{j=1}^t I_j^{PLC},$$

where $NI_1^{PLC} = 0$. Similar to the WMT scheme, the PLC scheme also has a final settlement as follows,

$$-1_{\{\tau \leq D\}}1_{\{NI_{\tau}^{PLC} < 0\}}NI_{\tau}^{PLC} - 1_{\{S_{D+1} > 0\}}1_{\{NI_D^{PLC} > 0\}}NI_D^{PLC},$$

where $\tau = \min\{t : S_{t+1} = 0\}$ is the finishing time of the whole task.

The decision problem under the PLC scheme, (P_t^{PLC}) , is then formulated as follows.

When $t < D$,

$$\begin{aligned} \min_{e_t, \dots, e_{\tau}} E_{t, S_t, NI_t^{PLC}} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma \beta \sum_{k=D}^{\tau} \delta^{k-t} u_2(S_{k+1}) \right. \\ & - I_t^{PLC} + 1_{\{S_{t+1}=0\}}1_{\{NI_t^{PLC} < 0\}}NI_t^{PLC} \\ & - \beta \sum_{k=t+1}^{\tau} \delta^{k-t} I_k^{PLC} + \beta \sum_{k=t+1}^D \delta^{k-t} 1_{\{S_{k+1}=0\}}1_{\{NI_k^{PLC} < 0\}}NI_k^{PLC} \\ & \left. + \beta \delta^{D-t} 1_{\{S_{D+1}>0\}}1_{\{NI_D^{PLC}>0\}}NI_D^{PLC} \right] \\ \text{s.t. } S_{j+1} &= \max\{S_j - Q_j e_j, 0\}, \quad NI_{j+1}^{PLC} = NI_j^{PLC} + I_j^{PLC}, \\ e_j &\in \{1, 2, \dots, e^u\}, \quad j = t, \dots, \tau. \end{aligned}$$

When $t = D$,

$$\begin{aligned} \min_{e_t, \dots, e_\tau} E_{t, S_t, NI_t^{PLC}} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma u_2(S_{k+1}) \right. \\ & + \gamma \beta \sum_{k=D}^{\tau} \delta^{k-t} u_2(S_{k+1}) - I_t^{PLC} + 1_{\{S_{t+1}=0\}} 1_{\{NI_t^{PLC} < 0\}} NI_t^{PLC} \\ & \left. + 1_{\{S_{t+1}>0\}} 1_{\{NI_t^{PLC} > 0\}} NI_t^{PLC} - \beta \sum_{k=t+1}^{\tau} \delta^{k-t} I_k^{PLC} \right] \\ \text{s.t. } S_{j+1} &= \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \quad j = t, \dots, \tau. \end{aligned}$$

When $t > D$,

$$\begin{aligned} \min_{e_t, \dots, e_\tau} E_{t, S_t, NI_t^{PLC}} & \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(e_k) + \gamma u_2(S_{t+1}) \right. \\ & \left. + \gamma \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_2(S_{k+1}) - I_t^{PLC} - \beta \sum_{k=t+1}^{\tau} \delta^{k-t} I_k^{PLC} \right] \\ \text{s.t. } S_{j+1} &= \max \{S_j - Q_j e_j, 0\}, \quad e_j \in \{1, 2, \dots, e^u\}, \quad j = t, \dots, \tau. \end{aligned}$$

The decisions applied by the task operator under the PLC scheme, $e_t^{PLC}(S_t, NI_t^{PLC})$, and the completion time of the task under the PLC scheme can be derived by Algorithm 1 and Algorithm 2 with $H(e_t; S_t, t)$ and $J(S_t, t)$ replaced (respectively) by

$$\begin{aligned} H_{PLC}(e_t; S_t, NI_t^{PLC}, t) &= g(e_t; S_t, t) + E_{t, S_t, NI_t^{PLC}} [h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t)] \\ &+ \beta \delta E_{t, S_t, NI_t^{PLC}} [J_{PLC}(S_{t+1}, NI_t^{PLC}, t+1)], \\ J_{PLC}(S_t, NI_t^{PLC}, t) &= \min_{e_t \in \{1, 2, \dots, e^u\}} \{g(e_t; S_t, t) + E_{t, S_t, NI_t^{PLC}} [h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t)] \\ &+ \delta E_{t, S_t, NI_t^{PLC}} [J_{PLC}(S_{t+1}, NI_t^{PLC}, t+1)]\}, \end{aligned}$$

where $J_{PLC}(0, NI_t^{PLC}, t) = 0$ and

$$h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t) = \begin{cases} -I_t^{PLC} + 1_{\{S_{t+1}=0\}} 1_{\{NI_t^{PLC} < 0\}} NI_t^{PLC}, & t < D, \\ -I_t^{PLC} + 1_{\{S_{t+1}=0\}} 1_{\{NI_t^{PLC} < 0\}} NI_t^{PLC} \\ \quad + 1_{\{S_{t+1}>0\}} 1_{\{NI_t^{PLC} > 0\}} NI_t^{PLC}, & t = D, \\ -I_t^{PLC}, & t > D. \end{cases}$$

A.2. Comparing WMT with PLC

Table A.1 compares the performance among the WMT, PLC, and LC schemes. Following a similar computational design to that in Section 5.2, we set $e^u = 12$, $q_i = i/4$ ($i = 1, \dots, 7$), $p_i = 1/7$ ($i = 1, \dots, 7$), $u_1(x) = x^2$, $u_2(x) = x^2$, $W = 48$, $\delta = 0.995$, $D = 6$, and $\gamma = 10$. In panel A, the expected tardiness (Tard.), the expected bonus (bonus), and the expected penalty (penalty) under different schemes when achieving the same probability of on-time completion (Prob.) are reported. In panel B, the probability of on-time completion (Prob.), the expected bonus (bonus), and the expected penalty (penalty) under different schemes when achieving the same expected tardiness (Tard.) are reported. In each panel,

the improvement ratio of WMT over PLC and the improvement ratio of PLC over LC are also reported.

As shown in Table A.1, the PLC scheme has better performance than the LC scheme, as expected. However, our WMT scheme also provides much better performance than PLC, especially when β is small. Moreover, when $\beta = 0$, both the LC and PLC schemes totally fail. The notation “—” means that no matter how large (or small) the bonus and the penalty are, the corresponding probability of on-time completion or expected tardiness cannot be achieved. The limitations of LC and PLC for $\beta = 0$ demonstrate that our WMT scheme effectively compensates for present bias. At the other extreme, when $\beta = 1$ (i.e., the task operator does not suffer from any time inconsistency), the classical LC works best, which demonstrates its popularity in common practice. But when the task operator suffers from present bias that is severe or even at the moderate level, WMT is a more efficient scheme.

A.3. Understanding the Strength of WMT

The results in Table A.1 show that our WMT scheme still performs much better than period-LC. There are two big advantages of the WMT scheme relative to classical LC: (i) WMT spends the bonus/penalty more accurately and efficiently than the LC scheme and (ii) by frequent monitoring of project progress S_t , WMT allocates the lump-sum bonus/penalty in classical LC across the whole project. Implementing LC in each period reduces the second advantage but does not affect the first one. The following analysis demonstrates that both period-LC and classical LC provide inefficient incentive functions, especially when the task operator suffers from severe present bias.

To be a good incentive scheme, the penalty function should be an increasing (or at least nondecreasing) function of S_{t+1} , which means that the larger the unfinished workload, the greater the penalty. However, we show in the next proposition and Figure A.1 that the current penalty function under PLC does not always work efficiently, especially when the task is already behind schedule or delayed. In contrast, the current penalty function of the WMT scheme is an increasing function of S_{t+1} (see Figure A.2), which always works efficiently, as shown in Lemma 1 in the paper.

Proposition A.1. *The current penalty function of PLC, $h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t)$, is a left-continuous step function, which has the following properties:*

Case I: When $t = 1$, or $D > t > 1$ and $S_t \leq S_t^{tar-PLC}$ (i.e., the task is on schedule), the current penalty function is a nonpositive and non-decreasing function of S_{t+1} .

Case II: When $D > t > 1$ and $S_t > S_t^{tar-PLC}$ (i.e., the task is behind schedule but not yet delayed), the current penalty function is a non-increasing function of S_{t+1} .

Case III: When $t \geq D$ and $S_t > 0$ (i.e., the task is already delayed), the current penalty function is a nonnegative and nonincreasing function of S_{t+1} .

Proposition A.1 reveals the major weakness of the PLC scheme, that is, the current penalty function does not always encourage harder work when the task is behind schedule or already delayed. These cases are especially

Table A.1. Performance Comparison Among WMT, PLC, and LC Schemes for a Single Task

Penal A: Performance comparison for same probability of on-time completion																
β	Prob.	WMT			PLC			Imp. (WMT to PLC)			LC			Imp. (PLC to LC)		
		Tard.	Bonus	Penalty	Tard.	Bonus	Penalty	Tard.	Bonus	Penalty	Tard.	Bonus	Penalty	Tard.	Bonus	Penalty
1	0.75	0.3673	15.4	7.6	0.3658	1.4	0.8	-0.4%	-1031.6%	-804.3%	0.3414	6.2	6.1	-7.2%	78.2%	86.2%
	0.8	0.2958	29.2	9.1	0.2863	17.2	7.1	-3.3%	-69.5%	-29.3%	0.2655	16.0	8.7	-7.9%	-8.0%	18.4%
	0.85	0.2248	59.7	13.1	0.2054	45.9	15.9	-9.5%	-30.0%	17.6%	0.1963	35.8	9.9	-4.6%	-28.3%	-61.1%
	0.9	0.1451	140.0	20.6	0.1339	130.0	34.2	-8.4%	-7.7%	39.7%	0.1276	86.4	10.2	-4.9%	-50.4%	-235.0%
0.6	0.75	0.3797	31.5	24.2	0.3805	31.1	26.5	0.2%	-1.3%	8.9%	0.4124	36.3	37.1	7.7%	14.2%	28.5%
	0.8	0.3031	53.8	24.0	0.3044	64.4	34.0	0.4%	16.4%	29.3%	0.3235	74.6	40.4	5.9%	13.7%	16.0%
	0.85	0.2201	89.6	21.3	0.2293	107.9	39.3	4.0%	17.0%	45.8%	0.2460	138.6	40.8	6.8%	22.2%	3.8%
	0.9	0.1438	164.3	22.2	0.1550	201.4	53.7	7.2%	18.4%	58.6%	0.1630	206.7	31.4	4.9%	2.6%	-70.8%
0.2	0.75	0.3811	50.1	55.8	0.5980	116.7	233.9	36.3%	57.1%	76.1%	0.8609	313.7	558.2	30.5%	62.8%	58.1%
	0.8	0.3017	83.0	49.9	0.4719	207.0	183.7	36.1%	59.9%	72.8%	0.6912	455.8	510.5	31.7%	54.6%	64.0%
	0.85	0.2265	139.1	40.6	0.3595	346.6	181.6	37.0%	59.9%	77.7%	0.5186	670.4	448.3	30.7%	48.3%	59.5%
	0.9	0.1494	210.2	29.5	0.2709	666.2	246.6	44.8%	68.5%	88.0%	0.3372	961.6	284.6	19.7%	30.7%	13.4%
0	0.75	0.3953	83.6	108.4	—	—	—	—	—	—	—	—	—	—	—	—
	0.8	0.3152	118.8	85.1	—	—	—	—	—	—	—	—	—	—	—	—
	0.85	0.2358	167.3	61.8	—	—	—	—	—	—	—	—	—	—	—	—
	0.9	0.1540	241.7	37.8	—	—	—	—	—	—	—	—	—	—	—	—
Penal B: Performance comparison for same expected tardiness																
β	Tard.	WMT			PLC			Imp. (WMT to PLC)			LC			Imp. (PLC to LC)		
		Prob.	Bonus	Penalty	Prob.	Bonus	Penalty	Prob.	Bonus	Penalty	Prob.	Bonus	Penalty	Prob.	Bonus	Penalty
1	0.45	0.7006	6.0	4.0	—	—	—	—	—	—	0.6895	1.5	2.7	—	—	—
	0.35	0.7616	18.0	8.1	0.7612	3.1	1.5	0.0%	-474.4%	-432.1%	0.7447	5.6	5.9	2.2%	44.0%	74.0%
	0.25	0.8323	45.2	11.2	0.8214	26.5	10.1	1.3%	-70.9%	-10.1%	0.8108	19.0	9.0	1.3%	-39.4%	-12.9%
	0.15	0.8963	130.7	19.8	0.8887	98.5	27.8	0.9%	-32.6%	29.0%	0.8844	65.6	10.3	0.5%	-50.3%	-170.4%
0.6	0.45	0.7020	17.0	22.3	0.7044	11.8	16.0	-0.3%	-44.6%	-40.0%	0.7248	25.9	33.8	-2.8%	54.5%	52.7%
	0.35	0.7702	39.1	24.2	0.7706	44.6	30.5	-0.1%	12.2%	20.5%	0.7852	60.5	39.7	-1.9%	26.3%	23.2%
	0.25	0.8320	97.5	22.9	0.8356	93.9	37.8	-0.4%	-3.9%	39.5%	0.8479	135.3	41.2	-1.4%	30.6%	8.2%
	0.15	0.8956	181.2	21.8	0.9033	214.8	56.4	-0.9%	15.7%	61.4%	0.9073	217.6	29.0	-0.4%	1.3%	-94.5%
0.2	0.45	0.7119	35.3	61.7	0.8091	227.6	177.3	-12.0%	84.5%	65.2%	0.8704	722.2	358.3	-7.0%	68.5%	50.5%
	0.35	0.7676	65.2	55.9	0.8547	365.7	185.9	-10.2%	82.2%	69.9%	0.8979	870.9	259.7	-4.8%	58.0%	28.4%
	0.25	0.8348	122.0	46.0	0.9087	738.0	248.5	-8.1%	83.5%	81.5%	0.9260	1,233.5	224.5	-1.9%	40.2%	-10.7%
	0.15	0.8996	213.5	32.1	—	—	—	—	—	—	—	—	—	—	—	—
0	0.45	0.7165	64.7	123.2	—	—	—	—	—	—	—	—	—	—	—	—
	0.35	0.7781	101.9	95.4	—	—	—	—	—	—	—	—	—	—	—	—
	0.25	0.8411	157.6	65.9	—	—	—	—	—	—	—	—	—	—	—	—
	0.15	0.9025	248.4	37.8	—	—	—	—	—	—	—	—	—	—	—	—

Note. Tard., tardiness; Prob., probability; Imp., improvement ratio.

important when the task operator suffers from severe present bias because severe present bias usually drives the task operator into Case II or Case III. Next, we further demonstrate the relative efficiency of the WMT and PLC schemes through an example.

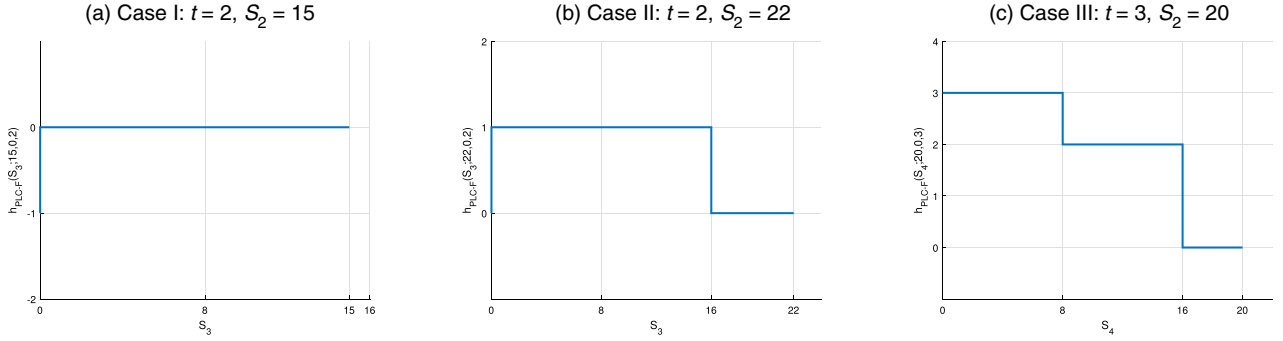
Example A.1. We set $S_1 = 24$, $D = 3$. We study the current penalty functions in different cases of Proposition A.1: (i) Case I, $t = 2$, $S_2 = 15$; (ii) Case II, $t = 2$, $S_2 = 22$; (iii) Case III, $t = 3$, $S_3 = 20$. First, we consider the PLC scheme with $B = P = 1$. The task progress targets are $S_2^{tar-PLC} = 16$, $S_3^{tar-PLC} = 8$, $S_4^{tar-PLC} = 0$. It is easy to compute that the cumulative incentives for different cases are all zero. According to the formulation of $h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t)$, we have

$$h_{PLC}(S_3; 15, 0, 2) = -1_{\{S_3=0\}}, \quad h_{PLC}(S_3; 22, 0, 2) = 1_{\{0 < S_3 \leq 16\}}, \\ h_{PLC}(S_4; 20, 0, 3) = 3 \times 1_{\{0 \leq S_4 \leq 8\}} + 2 \times 1_{\{8 \leq S_4 \leq 16\}}.$$

Figure A.1 shows the current penalty functions for different cases under the PLC scheme. We observe that for Cases II and III, the current penalty functions under PLC are clearly decreasing functions, which do not encourage harder work in the sense that a larger effort may result in a larger penalty. Furthermore, although the current penalty functions for Case I are nondecreasing, they do not provide any bonus for a relatively large range of the state, for example, $(0, 15]$ for that case.

Now, we consider the WMT scheme. The task progress target can be updated according to Equation (5) in the paper. For the three cases studied in this example, only the task progress target S_3^{target} in Case I, $t = 2$, $S_2 = 15$ is reduced to seven. Assuming $S_2 = 22$, the cumulative incentives for the three cases are 1, -6, and -18, respectively. According to the formulation of $h_{WMT}(S_{t+1}; S_t, NI_t, t)$, we

Figure A.1. (Color online) Current Penalty Functions of the PLC Scheme for Different Cases



have

$$\begin{aligned} h_{WMT}(S_3; 15, 1, 2) &= S_3 - 7, \\ h_{WMT}(S_3; 22, -6, 2) &= (S_3 - 8)1_{\{0 < S_3\}} - 14 \times 1_{\{S_3=0\}}, \\ h_{WMT}(S_4; 20 - 18, 3) &= S_4 1_{\{0 < S_4\}} - 18 \times 1_{\{S_4=0\}}. \end{aligned}$$

Figure A.2 shows the current penalty functions for different cases under the WMT scheme. We observe that the current penalty functions are all strictly increasing functions of S_{t+1} , which makes our WMT scheme a more efficient scheme.

Appendix B. Discussion for Partially Naive and Sophisticated Task Operators

Recall that depending on how the task operators perceive their future present bias level and how they balance current and future decisions, the task operators can be classified as naive, partially naive, and sophisticated (O'Donoghue and Rabin 2001). In this section, we provide a unified formulation for all three types of task operators. Define the perceived future preference for future time $s > t$ as,

$$U_s^t(u_s, u_{s+1}, \dots, u_T) = u_s + \hat{\beta} \sum_{j=s+1}^T \delta^{j-s} u_j,$$

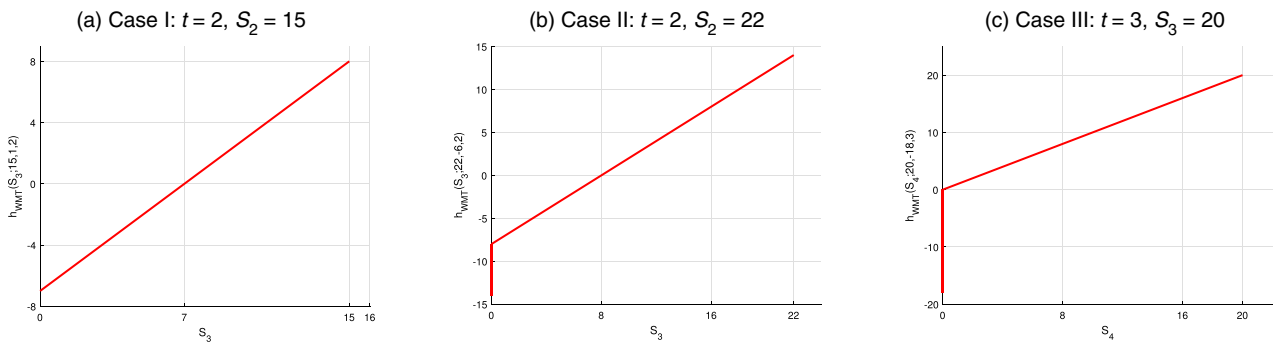
where $\hat{\beta}$ represents the perceived future present bias level of the task operators instead of their real present bias level β . The naive type task operators naively believe that their future selves will not suffer from any present bias in the

future, that is, $\hat{\beta} = 1$. The sophisticated task operators fully realize that they will suffer from the same degree of present bias in the future, that is, $\hat{\beta} = \beta$. The partially naive task operators lie in the middle in that they believe that their future selves will suffer from present bias but at a lower level than their real present bias level, that is, $\hat{\beta} > \beta$.

By introducing the perceived level of future present bias level $\hat{\beta}$, we can also unify the decision problems of all three types of task operators into a single framework. Following the sequential intrapersonal game literature (Grenadier and Wang 2007, Basak and Chabakauri 2010), the partially naive and sophisticated task operators face the following nested decision problem,

$$\begin{aligned} (NP_t) \min_{e_t} & \begin{cases} E_{t,S_t} \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(\hat{e}_k) + \gamma \beta \sum_{k=D}^{\tau} \delta^{k-t} u_2(S_{k+1}) \right], & j < D \\ E_{t,S_t} \left[u_1(e_t) + \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_1(\hat{e}_k) + u_2(S_{t+1}) \right. \\ \quad \left. + \gamma \beta \sum_{k=t+1}^{\tau} \delta^{k-t} u_2(S_{k+1}) \right], & j \geq D \end{cases} \\ \text{s.t. } & S_{t+1} = \max \{S_t - Q_t e_t, 0\}, \\ & S_{j+1} = \max \{S_j - Q_j \hat{e}_j, 0\}, \quad \hat{e}_j \in \{1, 2, \dots, e^u\}, \\ & \quad j = t+1, \dots, \tau, \\ & \hat{e}_j \text{ solves problem } (\widehat{NP}_j), \quad j = t+1, \dots, \tau, \end{aligned}$$

Figure A.2. (Color online) Current Penalty Functions of the WMT Scheme for Different Cases



where NP in (NP_t) and $(\widehat{NP}_j)$ is the abbreviation of nested problem, and the perceived future decision problem $(\widehat{NP}_j)$ is given as follows,

$$(\widehat{NP}_j) \min_{e_j} \begin{cases} E_{j,S_j} \left[u_1(e_j) + \hat{\beta} \sum_{k=j+1}^{\tau} \delta^{k-j} u_1(\hat{e}_k) + \gamma \hat{\beta} \sum_{k=D}^{\tau} \delta^{k-j} u_2(S_{k+1}) \right], & j < D \\ E_{j,S_j} \left[u_1(e_j) + \hat{\beta} \sum_{k=j+1}^{\tau} \delta^{k-j} u_1(\hat{e}_k) + \gamma u_2(S_{j+1}) \right. \\ \quad \left. + \gamma \hat{\beta} \sum_{k=j+1}^{\tau} \delta^{k-j} u_2(S_{k+1}) \right], & j \geq D \end{cases}$$

$$\text{s.t. } S_{j+1} = \max \{S_j - Q_j e_j, 0\},$$

$$S_{\ell+1} = \max \{S_{\ell} - Q_{\ell} \hat{e}_{\ell}, 0\},$$

$$\hat{e}_{\ell} \in \{1, 2, \dots, e^u\}, \quad \ell = j+1, \dots, \tau,$$

$$\hat{e}_{\ell} \text{ solves problem } (\widehat{NP}_{\ell}), \quad \ell = j+1, \dots, \tau.$$

More intuitively, at time t , the task operator plays a sequential intrapersonal game with his or her future self by believing his or her future self will make and follow the decisions in the future periods with perceived future present bias level $\hat{\beta}$; that is, $\hat{e}_j$ solves problems $(\widehat{NP}_j)$, $j \geq t+1$. When $\hat{\beta} = 1$, the nested decision problem (NP_t) reduces into the simple version (P_t) , that is, the decision problem for naive type task operators, as discussed in Section 2.1 of the paper.

Endnotes

¹ In Lemma 1, we omit the effect of δ by setting $\delta = 1$. Following the same procedure, we find that Lemma 1(a) still holds when $\delta < 1$ and Lemma 1(b) holds when $\delta < (T-D-1)/(T-D)$. In practice, condition $\delta < (T-D-1)/(T-D)$ typically holds.

² The values in the bonus, penalty and tardiness columns are rounded to one decimal place; however, the improvement ratio is computed based on exact values.

³ Note that, if the suggested adjustment one-fourth is changed to any fixed proportional number, such as $1/a$, our main results still hold.

⁴ All three intervals of β for the three cases in Example 1 include 0.4.

⁵ The two intervals of β for Cases II and III in Example 1 include 0.6.

⁶ Because of space limitation, we only perform this analysis at the task level. However, the relative strength of WMT still continues at the project level. The reason is that both classical and period-LC perform inefficiently when the task is already delayed or at least behind schedule (Case II and Case III as discussed in Section A.3). In a project with many tasks, their delay accumulates, which makes Case II and Case III more likely to occur.

⁷ Note that the progress target is fixed in that the manager divides the task into subtasks and sets the progress target without changing it during the project. If we allow the project manager to change the progress target according to the same target setting formula (5) as for the WMT scheme, the WMT scheme still performs much better than period-LC.

References

- Abu-Hilje SF, Ibbs CW (1989) Schedule-based construction incentives. *J. Construction Engrg. Management* 115(3):430–443.
- Akerlof G (1991) Procrastination and obedience. *Amer. Econom. Rev.* 81(2):1–19.

- Arditi D, Yasamis F (1998) Incentive/disincentive contracts: Perceptions of owners and contractors. *J. Construction Engrg. Management* 124(5):361–373.
- Basak S, Chabakauri G (2010) Dynamic mean-variance asset allocation. *Rev. Financial Stud.* 23(8):2970–3016.
- Bayiz M, Corbett CJ (2005) Coordination and incentive contracts in project management under asymmetric information. Working paper, Anderson Graduate School of Management, University of California, Los Angeles.
- Bertsimas D, Natarajan K, Teo C-P (2004) Probabilistic combinatorial optimization: Moments, semidefinite programming, and asymptotic bounds. *SIAM J. Optim.* 15(1):185–209.
- Bower D, Ashby G, Gerald K, Smyk W (2002) Incentive mechanisms for project success. *J. Management Engrg.* 18(1):37–43.
- Chen B, Hall NG (2021) Incentive schemes for resolving Parkinson's Law in project management. *Eur. J. Oper. Res.* 288(2):666–681.
- Corcione A (2019) Workplace incentives that your employees want. Accessed August 29, 2022, <https://www.businessnewsdaily.com/9240-engaging-employee-incentives.html>.
- Della Vigna S, Lindner A, Reizer B, Schmieder JF (2017) Reference-dependent job search: Evidence from Hungary. *Quart. J. Econom.* 132(4):1969–2018.
- Demeulemeester E, Vanhoucke M, Herroelen W (2003) RanGen: A random network generator for activity-on-the-node networks. *J. Scheduling* 6(1):17–38.
- Ericson KM (2017) On the interaction of memory and procrastination: Implications for reminders, deadlines, and empirical estimation. *J. Eur. Econom. Assoc.* 15(3):692–719.
- Fotsch B, Case J (2015) How to build incentive plans that actually work. *Forbes* (August 24), accessed October 3, 2022, <https://www.forbes.com/sites/fotschcase/2015/08/24/incentive-plans-that-actually-work/#559c0e9061e6>.
- Grenadier SR, Wang N (2007) Investment under uncertainty and time-inconsistent preferences. *J. Financial Econom.* 84(1):2–39.
- Gutierrez GJ, Kouvelis P (1991) Parkinson's Law and its implications for project management. *Management Sci.* 37(8):990–1001.
- Hall NG (2016) Research and teaching opportunities in project management. *Tutorials in Operations Research* (INFORMS, Catonsville, MD).
- Howard WE, Bell LC, McCormick RE (1997) Economic principles of contractor compensation. *J. Manufacturing Engrg.* 13(5):81–89.
- Hu X, Cui N, Demeulemeester E (2015) Effective expediting to improve project due date and cost performance through buffer management. *Internat. J. Production Res.* 53(5):1460–1471.
- Kaur S, Kremer M, Mullainathan S (2015) Self-control at work. *J. Political Econom.* 123(6):1227–1277.
- Kerzner H (2017) *Project Management: A Systems Approach to Planning, Scheduling, and Controlling* (John Wiley & Sons, New York).
- Khang DB, Myint YM (1999) Time, cost and quality trade-off in project management: A case study. *Internat. J. Project Management* 17(4):249–256.
- Klastorin T (2012) *Project Management: Tools and Tradeoffs*, 3rd ed. (Pearson, Upper Saddle River, NJ).
- Krakowski M (1974) PERT and Parkinson's Law. *Interfaces* 5(1):35–40.
- Laibson D (1994) Essays in hyperbolic discounting. Unpublished doctoral thesis, Massachusetts Institute of Technology, Cambridge, MA.
- Lewis G, Bajari P (2014) Moral hazard, incentive contracts, and risk: Evidence from procurement. *Rev. Econom. Stud.* 81(3):1201–1228.
- Li Q, Nasiry J, Yu P (2017) Optimal stopping under present-biased preferences. Working paper, Fudan University, Shanghai, China.
- Liao CN, Chen YJ (2021) Design of long term conditional cash transfer program to encourage healthy habits. *Production Oper. Management* 30(11):3987–4003.

- Meier S, Sprenger CD (2015) Temporal stability of time preferences. *Rev. Econom. Statist.* 97(2):273–286.
- O'Donoghue T, Rabin M (1999a) Doing it now or later. *Amer. Econom. Rev.* 89(1):103–124.
- O'Donoghue T, Rabin M (1999b) Incentives for procrastinators. *Quart. J. Econom.* 114(3):769–816.
- O'Donoghue T, Rabin M (2001) Choice and procrastination. *Quart. J. Econom.* 116(1):121–160.
- Parkinson CN (1955) Parkinson's Law. *The Economist* (November 19), <https://www.economist.com/news/1955/11/19/parkinsons-law>.
- Parkinson CN (1958) *Parkinson's Law: The Pursuit of Progress* (John Murray, London).
- Paserman MD (2008) Job search and hyperbolic discounting: Structural estimation and policy evaluation. *Econom. J. (London)* 118(531):1418–1452.
- Phelps ES, Pollak R (1968) On second-best national saving and game-equilibrium growth. *Rev. Econom. Stud.* 35(2):185–199.
- Project Management Institute (2013) *A Guide to the Project Management Body of Knowledge (PMI Guide)* (PMI Publications, Newton Square, PA).
- Rose TM, Manley K (2011) Motivation toward financial incentive goals on construction projects. *J. Bus. Res.* 64(7):765–773.
- Standish Group (2015) *CHAOS Report*. Accessed August 29, 2022, https://www.standishgroup.com/sample_research_files/CHAOSReport2015-Final.pdf.
- Thaler RH, Benartzi S (2004) Save more tomorrowTM: Using behavioral economics to increase employee saving. *J. Political Econom.* 112(S1):S164–S187.
- Williams TM (1995) What are PERT estimates? *J. Oper. Res. Soc.* 46(12):1498–1504.
- Wu Y, Krishnan V, Ramachandran K (2014) Managing cost salience and procrastination in projects: Compensation and team composition. *Production Oper. Management* 23(8):1299–1311.
- Wysocki RK (2009) *Effective Project Management*, 5th ed. (John Wiley & Sons, Indianapolis).

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“Work More Tomorrow: Resolving Present Bias in Project Management”

Online Appendix: Proofs

This document contains all the proofs in the paper “Work More Tomorrow: Resolving Present Bias in Project Management”.

1 Proof of Lemma 1

Proof. (a) It is easy to check that $-I_t$ is a strictly increasing function of S_{t+1} , $1_{\{S_{t+1}=0\}}1_{\{NI_t<0\}}NI_t$, and $1_{\{S_{t+1}=0\}}1_{\{NI_t<0\}}NI_t+1_{\{S_{t+1}>0\}}1_{\{NI_t>0\}}NI_t$ are nondecreasing step functions of S_{t+1} , which implies that $h_{WMT}(S_{t+1}; S_t, NI_t, t)$ is a strictly increasing function of S_{t+1} . As S_{t+1} is a non-increasing function of e_t , we further conclude that $E_{t,S_t,NI_t}[h_{WMT}(S_{t+1}; S_t, NI_t, t)]$ is a nonincreasing function of e_t .

When $t > D$, (P_t^{WMT}) does not involve NI_t and $J_{WMT}(S_t, NI_t, t)$ is a constant function of NI_t . By backward induction, we assume that $J_{WMT}(S_{t+1}, NI_{t+1}, t+1)$ is a nondecreasing function of NI_{t+1} and prove that $J_{WMT}(S_t, NI_t, t)$ is a nondecreasing function of NI_t . Suppose we have two states of NI_t at time t , $NI_t^2 > NI_t^1$. Then, for any effort level e_t and any state S_t , we have

$$\begin{aligned} & g(e_t; S_t, t) + E_{t,S_t,NI_t^2}[h_{WMT}(S_{t+1}; S_t, NI_t^2, t)] + E_{t,S_t,NI_t^2}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)] \\ & \geq g(e_t; S_t, t) + E_{t,S_t,NI_t^1}[h_{WMT}(S_{t+1}; S_t, NI_t^1, t)] + E_{t,S_t,NI_t^1}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]. \end{aligned}$$

Since $J_{WMT}(S_{t+1}, NI_{t+1}, t+1)$ is a nondecreasing function of NI_{t+1} , NI_{t+1} is an increasing function of NI_t , and $E_{t,S_t,NI_t}[h_{WMT}(S_{t+1}; S_t, NI_t, t)]$ is a nondecreasing function of NI_t , the inequality holds, which further implies $J_{WMT}(S_t, NI_t, t)$ is a nondecreasing function of NI_t .

Next, we prove that when $t > D$, $J_{WMT}(S_t, NI_t, t)$ is an increasing function of S_t by backward induction. Since the smallest effort level is 1 and $\min_i\{m_i\} = 1$, the maximum period to finish the task is $T = WM$. At time $t = T$, the only possible states of S_T are

$S_T = 1/M$ and $S_T = 0$. Since $J_{WMT}(0, NI_{T+1}, T+1) = 0$, it is easy to show that

$$\begin{aligned} & J_{WMT}(1/M, NI_T, T) \\ &= \min_{e_T} g(e_T; 1/M, T) + E_{T, 1/M}[h_{WMT}(S_{T+1}; 1/M, NI_T, T)] = u_1(1) > 0 = J_{WMT}(0, NI_T, T). \end{aligned}$$

Therefore, $J_{WMT}(S_T, NI_T, T)$ is an increasing function of S_T . We assume that for time $t+1$, $J_{WMT}(S_{t+1}, NI_{t+1}, t+1)$ is an increasing function of S_{t+1} and prove that it is still an increasing function of S_t for time t . Suppose we have two states of S_t at time t , $S_t^2 > S_t^1 \geq 0$, and the optimal effort levels for the auxiliary optimization problem under the WMT scheme are e_t^2 and e_t^1 , then

$$\begin{aligned} & J_{WMT}(S_t^2, NI_t, t) \\ &= \{g(e_t; S_t^2, t) + E_{t, S_t^2, NI_t}[h_{WMT}(S_{t+1}; S_t^2, NI_t, t)] + E_{t, S_t^2, NI_t}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]\}_{\{e_t=e_t^2\}} \\ &> \{g(e_t; S_t^1, t) + E_{t, S_t^1, NI_t}[h_{WMT}(S_{t+1}; S_t^1, NI_t, t)] + E_{t, S_t^1, NI_t}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]\}_{\{e_t=e_t^2\}} \\ &\geq \{g(e_t; S_t^1, t) + E_{t, S_t^1, NI_t}[h_{WMT}(S_{t+1}; S_t^1, NI_t, t)] + E_{t, S_t^1, NI_t}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]\}_{\{e_t=e_t^1\}} \\ &= J_{WMT}(S_t^1, NI_t, t). \end{aligned}$$

Since $J_{WMT}(S_{t+1}, NI_{t+1}, t+1)$ is an increasing function of S_{t+1} and a constant function of NI_{t+1} , and $E_{t, S_t, NI_t}[h_{WMT}(S_{t+1}; S_t, NI_t, t)]$ is a nondecreasing function of S_t , the first inequality holds. Since e_t^1 is the optimal effort level for the auxiliary optimization problem with state S_t^1 , the second inequality holds. Thus, $J_{WMT}(S_t, NI_t, t)$ is an increasing function of S_t . As S_{t+1} is a nonincreasing function of e_t , $E_{t, S_t, NI_t}[J_{WMT}(S_{t+1}, NI_{t+1}, t+1)]$ is a nonincreasing function of e_t .

(b) For the future penalty function, we also use backward induction to prove the statement. At time $t = T$, the only possible states are $S_T = 1/M$ and $S_T = 0$. Since $J_{LC}(0, T+1) = 0$, it is easy to show that

$$\begin{aligned} J_{LC}(1/M, T) &= \min_{e_T} g(e_T; 1/M, T) + E_{T, 1/M}[h_{LC}(S_{T+1}; 1/M, T)] = u_1(1) + h_{LC}(0; 1/M, T) \\ &> h_{LC}(0; 1/M, T) = P(T-D) > P(T-D-1) = h_{LC}(0; 1/M, T-1). \end{aligned}$$

Therefore, the statement holds at time T . We assume that the statement holds for time $t + 1$ and prove it for time t . Suppose we have two states at time t , $S_t^2 > S_t^1 > 0$. We define a new function

$$\tilde{J}_{LC}(S_{t+1}, t + 1; S_t, t) = J_{LC}(S_{t+1}, t + 1) + h_{LC}(S_{t+1}; S_t, t) = \begin{cases} J_{LC}(S_{t+1}, t + 1), & \text{if } S_{t+1} > 0, \\ h_{LC}(0; S_t, t), & \text{if } S_{t+1} = 0, \end{cases}$$

which is an increasing function of $S_{t+1} \geq 0$ due to backward induction. We further observe that, for any e_t ,

$$\begin{aligned} & E_{t, S_t^2}[h_{LC}(S_{t+1}; S_t^2, t)] + E_{t, S_t^2}[J_{LC}(S_{t+1}, t + 1)] = E_{t, S_t^2}[\tilde{J}_{LC}(S_{t+1}, t + 1; S_t^2, t)] \\ &= \sum_{i=1}^n p_i \tilde{J}_{LC}(\max\{S_t^2 - q_i e_t\}, t + 1; S_t^2, t) > \sum_{i=1}^n p_i \tilde{J}_{LC}(\max\{S_t^1 - q_i e_t\}, t + 1; S_t^1, t) \\ &= E_{t, S_t^1}[\tilde{J}_{LC}(S_{t+1}, t + 1; S_t^1, t)] = E_{t, S_t^1}[h_{LC}(S_{t+1}; S_t^1, t)] + E_{t, S_t^1}[J_{LC}(S_{t+1}, t + 1)]. \end{aligned}$$

Since $h_{LC}(0; S_t^2, t) = h_{LC}(0; S_t^1, t)$ and $\tilde{J}_{LC}(S_{t+1}, t + 1; S_t, t)$ is an increasing function of S_{t+1} , the inequality holds. Applying a similar argument for function $J_{WMT}(S_t, NI_t, t)$ being an increasing function of S_t , we have $J_{LC}(S_t^2, t) > J_{LC}(S_t^1, t)$. Finally, when $S_t = 1/M$, we have

$$\begin{aligned} J_{LC}(1/M, t) &= \min_{e_t} g(e_t; 1/M, t) + E_{t, 1/M}[h_{LC}(S_{t+1}; 1/M, t)] = u_1(1) + h_{LC}(0; 1/M, t) \\ &> h_{LC}(0; 1/M, t). \end{aligned}$$

When $t > D$, $h_{LC}(0; 1/M, t) = P(t - D) > P(t - D - 1) = h_{LC}(0; 1/M, t - 1)$. When $t \leq D$, $h_{LC}(0; 1/M, t) = -B(D - t) > -B(D - t + 1) = h_{LC}(0; 1/M, t - 1)$. Thus, $J_{LC}(S_t, t)$ is an increasing function of $S_t > 0$ and $J_{LC}(S_t, t) > h_{LC}(0; 1/M, t - 1)$. As S_{t+1} is a nonincreasing function of e_t , $E_{t, S_t}[J_{LC}(S_{t+1}, t + 1)]$ is a nonincreasing function of e_t .

For the current penalty function, we consider three cases:

Case I: When $t < D$ and $S_t \leq \max_i \{q_i e^u\}$, $h_{LC}(S_{t+1}; S_t, t) = -B(D - t)1_{\{S_{t+1}=0\}}1_{\{S_t>0\}}$, which is a nondecreasing step function of S_{t+1} . Furthermore, a large e_t implies a large probability of finishing the task in period t , which further implies small $E_{t, S_t}[h_{LC}(S_{t+1}; S_t, t)]$. Therefore, $E_{t, S_t}[h_{LC}(S_{t+1}; S_t, t)]$ is a nonincreasing function of e_t .

Case II: When $t < D$ and $S_t > \max_i \{q_i e^u\}$, $h_{LC}(S_{t+1}; S_t, t) = -B(D - t)1_{\{S_{t+1}=0\}}1_{\{S_t>0\}}$,

which is also a nondecreasing step function of S_{t+1} . However, the probability of finishing the task in period t is zero, which implies $E_{t,S_t}[h_{LC}(S_{t+1}; S_t, t)] = 0$ for all e_t .

Case III: When $t \geq D$, $h_{LC}(S_{t+1}; S_t, t) = P(t - D)1_{\{S_{t+1}=0\}}1_{\{S_t>0\}}$, which is a nonincreasing step function of S_{t+1} . Furthermore, a large e_t implies a small S_{t+1} and a large probability of finishing the task in period t . Therefore, $E_{t,S_t}[h_{LC}(S_{t+1}; S_t, t)]$ is a nondecreasing function of e_t . $\square$

2 Proof of Theorem 1

Proof. We first derive the if and only if condition for Case III, and then use the same procedure to obtain similar results for Case II and Case I.

For Case III, when $t > D$ and $S_t > 0$, $\tilde{J}_W(e_t)$ and $\tilde{J}_L(e_t)$ are both nonincreasing function of e_t . The inequality of $e_t^{WMT}(S_t, NI_t) \geq e_t^{LC}(S_t)$ is equivalent to

$$e_t^{WMT}(S_t, NI_t) \geq i \geq e_t^{LC}(S_t),$$

holds for some i . For a given i , we have

$$\begin{aligned} & e_t^{WMT}(S_t, NI_t) \geq i, \\ \Leftrightarrow & \tilde{g}(i) + \tilde{h}_W(i) + \beta\delta\tilde{J}_W(i) \leq \tilde{g}(e_t) + \tilde{h}_W(e_t) + \beta\delta\tilde{J}_W(e_t), \quad \text{for any } 1 \leq e_t < i, \\ \Leftrightarrow & \begin{cases} 1 \geq \beta \geq \max \left\{ \frac{\Delta\tilde{g}(i, e_t) + \Delta\tilde{h}_W(i, e_t)}{\delta\Delta\tilde{J}_W(e_t, i)}, 0 \right\}, & \text{if } \Delta\tilde{J}_W(e_t, i) > 0, \\ 1 \geq \beta \geq 0, & \text{if } \Delta\tilde{J}_W(e_t, i) = 0, \Delta\tilde{g}(i, e_t) \leq \Delta\tilde{h}_W(e_t, i), \\ \beta \in \emptyset, & \text{if } \Delta\tilde{J}_W(e_t, i) = 0, \Delta\tilde{g}(i, e_t) > \Delta\tilde{h}_W(e_t, i), \end{cases} \quad \text{for any } 1 \leq e_t < i \\ \Leftrightarrow & \underline{\beta}(i) \leq \beta \leq 1. \\ & i \geq e_t^{LC}(S_t), \\ \Leftrightarrow & \tilde{g}(i) + \tilde{h}_L(i) + \beta\delta\tilde{J}_L(i) \leq \tilde{g}(e_t) + \tilde{h}_L(e_t) + \beta\delta\tilde{J}_L(e_t), \quad \text{for any } e^u \geq e_t > i, \\ \Leftrightarrow & \begin{cases} 0 \leq \beta \leq \min \left\{ \frac{\Delta\tilde{g}(i, e_t) + \Delta\tilde{h}_L(i, e_t)}{\delta\Delta\tilde{J}_L(e_t, i)}, 1 \right\} & \text{if } \Delta\tilde{J}_L(e_t, i) < 0, \\ 0 \leq \beta \leq 1, & \text{if } \Delta\tilde{J}_L(e_t, i) = 0, \Delta\tilde{g}(i, e_t) \leq \Delta\tilde{h}_L(e_t, i), \\ \beta \in \emptyset, & \text{if } \Delta\tilde{J}_L(e_t, i) = 0, \Delta\tilde{g}(i, e_t) > \Delta\tilde{h}_L(e_t, i), \end{cases} \quad \text{for any } e^u \geq e_t > i \\ \Leftrightarrow & 0 \leq \beta \leq \bar{\beta}(i), \end{aligned}$$

where $\emptyset$ denotes the empty set. Thus, the if and only if condition of $e_t^{WMT}(S_t, NI_t) \geq i \geq e_t^{LC}(S_t)$ is $\underline{\beta}(i) \leq \beta \leq \bar{\beta}(i)$. By changing i from 1 to e^u , we obtain the conclusion.

For Case II, when $t \leq D$ and $S_t > \max_i \{q_i e^u\}$, $\tilde{J}_L(e_t)$ is nonincreasing function of e_t , $\tilde{h}_L(e_t)$ is a constant function of e_t , $\tilde{J}_W(e_t)$ may not a nonincreasing function of e_t . Therefore, for a given i , we have

$$\begin{aligned}
& e_t^{WMT}(S_t, NI_t) \geq i, \\
& \Leftrightarrow \tilde{g}(i) + \tilde{h}_W(i) + \beta \delta \tilde{J}_W(i) \leq \tilde{g}(e_t) + \tilde{h}_W(e_t) + \beta \delta \tilde{J}_W(e_t), \quad \text{for any } 1 \leq e_t < i, \\
& \Leftrightarrow \begin{cases} 1 \geq \beta \geq \max \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)}, 0 \right\}, & \text{if } \Delta \tilde{J}_W(e_t, i) > 0, \\ 1 \geq \beta \geq 0, & \text{if } \Delta \tilde{J}_W(e_t, i) = 0, \Delta \tilde{g}(i, e_t) \leq \Delta \tilde{h}_W(e_t, i), \\ \beta \in \emptyset, & \text{if } \Delta \tilde{J}_W(e_t, i) = 0, \Delta \tilde{g}(i, e_t) > \Delta \tilde{h}_W(e_t, i), \\ 0 \leq \beta \leq \min \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)}, 1 \right\}, & \text{if } \Delta \tilde{J}_W(e_t, i) < 0, \end{cases} \quad \text{for any } 1 \leq e_t < i, \\
& \Leftrightarrow \underline{\beta}(i) \leq \beta \leq \min \left\{ \min_{1 \leq e_t < i} \left\{ \frac{\Delta \tilde{g}(i, e_t) + \Delta \tilde{h}_W(i, e_t)}{\delta \Delta \tilde{J}_W(e_t, i)} 1_{\{\Delta \tilde{J}_W(e_t, i) < 0\}} + 1_{\{\Delta \tilde{J}_W(e_t, i) \geq 0\}} \right\}, 1 \right\}.
\end{aligned}$$

Applying the same argument on $i \geq e_t^{LC}(S_t)$ as in Case III, we obtain the conclusion.

For Case I, when $t \leq D$ and $S_t \leq \max_i \{q_i e^u\}$, $\tilde{J}_L(e_t)$ and $\tilde{h}_L(e_t)$ are nonincreasing functions of e_t , $\tilde{J}_W(e_t)$ may not be a nonincreasing function of e_t . We can use the same argument on $e_t^{WMT}(S_t, NI_t) \geq i$ and $i \geq e_t^{LC}(S_t)$ as in Case III and Case II to obtain the conclusion. $\square$

3 Proof of Theorem 2

Proof. For a serial network with two tasks and $s \geq D_1 + D_2$, we have

$$\begin{aligned}
Pr_{WMT}(\tau_1 + \tau_2 \leq s) &= \sum_{k=1}^{D_1} f_{WMT}^{1,D_1}(k) F_{WMT}^{2,D_2}(s-k) + \sum_{j=D_1+1}^{s-1} f_{WMT}^{1,D_1}(j) F_{WMT}^{2,D_2-1}(s-j), \\
Pr_{LC}(\tau_1 + \tau_2 \leq s) &= \sum_{k=1}^{D_1} f_{LC}^{1,D_1}(k) F_{LC}^{2,D_2}(s-k) + \sum_{j=D_1+1}^{s-1} f_{LC}^{1,D_1}(j) F_{LC}^{2,D_2-1}(s-j).
\end{aligned}$$

Then,

$$Pr_{WMT}(\tau_1 + \tau_2 \leq s) - Pr_{LC}(\tau_1 + \tau_2 \leq s)$$

$$\begin{aligned}
&\geq \sum_{k=1}^{D_1} [f_{WMT}^{1,D_1}(k) - f_{LC}^{1,D_1}(k)] F_{LC}^{2,D_2}(s-k) + \sum_{j=D_1+1}^{s-1} f_{WMT}^{1,D_1}(j) F_{WMT}^{2,D_2-1}(s-j) - f_{LC}^{1,D_1}(j) F_{LC}^{2,D_2-1}(s-j) \\
&= G(s) \\
&\geq 0.
\end{aligned}$$

$F_{WMT}^{2,D_2}(s-k) \geq F_{LC}^{2,D_2}(s-k)$, ($k = 1, \dots, D_1$) leads to the first inequality. Furthermore, based on the definition of expected tardiness, $E[(\tau_1 + \tau_2 - D_1 - D_2)1_{\{\tau_1 + \tau_2 \geq D_1 + D_2\}}] = \sum_{s=D_1+D_2}^T Pr(\tau_1 + \tau_2 \leq s)$, the WMT scheme has smaller expected tardiness than the LC scheme for the serial network. $\square$

4 Proof of Theorem 3

Proof. First, consider a serial network with two tasks, task 1 and task 2. We introduce a useful inequality. When $F_{WMT}^{1,D_1}(k) \geq F_{LC}^{1,D_1}(k)$, we have

$$\begin{aligned}
&[F_{WMT}^{1,D_1}(k) - F_{LC}^{1,D_1}(k)] F_{LC}^{2,D_2}(s-k) + [f_{WMT}^{1,D_1}(k+1) - f_{LC}^{1,D_1}(k+1)] F_{LC}^{2,D_2}(s-k-1) \\
&\geq [F_{WMT}^{1,D_1}(k+1) - F_{LC}^{1,D_1}(k+1)] F_{LC}^{2,D_2}(s-k-1).
\end{aligned} \tag{1}$$

When $s \leq D_1 + 1$, we have

$$\begin{aligned}
&Pr_{WMT}(\tau_1 + \tau_2 \leq s) - Pr_{LC}(\tau_1 + \tau_2 \leq s) \\
&= \sum_{k=1}^{s-1} f_{WMT}^{1,D_1}(k) F_{WMT}^{2,D_2}(s-k) - \sum_{k=1}^{s-1} f_{LC}^{1,D_1}(k) F_{LC}^{2,D_2}(s-k) \\
&\geq \sum_{k=1}^{s-1} [f_{WMT}^{1,D_1}(k) - f_{LC}^{1,D_1}(k)] F_{LC}^{2,D_2}(s-k) \\
&\geq [F_{WMT}^{1,D_1}(s-1) - F_{LC}^{1,D_1}(s-1)] F_{LC}^{2,D_2}(1) \\
&\geq 0,
\end{aligned}$$

where the fact of $F_{WMT}^{2,D_2}(s-k) \geq F_{LC}^{2,D_2}(s-k)$, ($k = 1, \dots, s-1$) leads to the first and last inequalities. Applying the above inequality (1) ($s-1$) times leads to the second inequality.

When $s \geq D_1 + 2$, by applying a similar discussion, we have

$$Pr_{WMT}(\tau_1 + \tau_2 \leq s) - Pr_{LC}(\tau_1 + \tau_2 \leq s)$$

$$\begin{aligned}
&= \sum_{k=1}^{D_1} f_{WMT}^{1,D_1}(k) F_{WMT}^{2,D_2}(s-k) - \sum_{k=1}^{D_1} f_{LC}^{1,D_1}(k) F_{LC}^{2,D_2}(s-k) \\
&\quad + \sum_{j=D_1+1}^{s-1} f_{WMT}^{1,D_1}(j) F_{WMT}^{2,D_2-1}(s-j) - \sum_{j=D_1+1}^{s-1} f_{LC}^{1,D_1}(j) F_{LC}^{2,D_2-1}(s-j) \\
&\geq \sum_{k=1}^{D_1} [f_{WMT}^{1,D_1}(k) - f_{LC}^{1,D_1}(k)] F_{LC}^{2,D_2}(s-k) + \sum_{j=D_1+1}^{s-1} [f_{WMT}^{1,D_1}(j) - f_{LC}^{1,D_1}(j)] F_{LC}^{2,D_2-1}(s-j) \\
&\geq [F_{WMT}^{1,D_1}(D_1) - F_{LC}^{1,D_1}(D_1)] F_{LC}^{2,D_2}(s-D_1) + \sum_{j=D_1+1}^{s-1} [f_{WMT}^{1,D_1}(j) - f_{LC}^{1,D_1}(j)] F_{LC}^{2,D_2-1}(s-j) \\
&\geq [F_{WMT}^{1,D_1}(D_1) - F_{LC}^{1,D_1}(D_1)] F_{LC}^{2,D_2-1}(s-D_1-1) + \sum_{j=D_1+1}^{s-1} [f_{WMT}^{1,D_1}(j) - f_{LC}^{1,D_1}(j)] F_{LC}^{2,D_2-1}(s-j) \\
&\geq [F_{WMT}^{1,D_1}(s-1) - F_{LC}^{1,D_1}(s-1)] F_{LC}^{2,D_2-1}(1) \\
&\geq 0.
\end{aligned}$$

Therefore, we have

$$Pr_{WMT}(\tau_1 + \tau_2 \leq t) \geq Pr_{LC}(\tau_1 + \tau_2 \leq t), \quad t = 1, 2, \dots, 2T,$$

which implies that the combined new task of task 1 and task 2 also satisfies the assumption for the first single task.

Next, we can consider the combination of task 1 and task 2 as the new first single task $\tilde{1}$. The serial network with three tasks is then reduced into a serial network with two tasks, $\tilde{1}$ and 3. Applying the above argument, we can conclude that the combined new task of $\tilde{1}$ and 3 also satisfies the assumption for the first single task. By induction, we would have

$$Pr_{WMT}(\sum_{i=1}^N \tau_i \leq t) \geq Pr_{LC}(\sum_{i=1}^N \tau_i \leq t), \quad t = 1, 2, \dots, NT,$$

which implies that the WMT scheme has a larger probability of on-time completion and a smaller expected tardiness than the LC scheme. $\square$

5 Proof of Proposition 1 in Appendix A3

Proof. As the current penalty function only depends on S_{t+1} through two indicator functions, $1_{\{S_{t+1} \leq S_{t+1}^{tar-PLC}\}}$ and $1_{\{S_{t+1}=0\}}$, the current penalty function is a left-continuous step function.

a) When $t = 1$, since $S_1 > S_{i+1}^{tar-PLC}$ and $NI_1^{PLC} = 0$, we have

$$h_{PLC}(S_2; S_1, NI_1^{PLC}, 1) = - \sum_{i=1}^D B(i-1) 1_{\{S_2 \leq S_{i+1}^{tar-PLC}\}},$$

which is a nonpositive and nondecreasing function of S_2 . Similarly, when $D > t > 1$ and $S_t \leq S_t^{tar-PLC}$, we have

$$h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t) = - \sum_{i=t}^D B(i-t) 1_{\{S_t > S_{i+1}^{tar-PLC}\}} 1_{\{S_{t+1} \leq S_{i+1}^{tar-PLC}\}} + 1_{\{S_{t+1}=0\}} 1_{\{NI_t^{PLC} < 0\}} NI_t^{PLC},$$

which is also a nonpositive and nondecreasing function of S_{t+1} .

b) When $D > t > 1$ and $S_t > S_t^{tar-PLC}$, we have

$$\begin{aligned} h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t) = & - \sum_{i=t}^D B(i-t) 1_{\{S_t > S_{i+1}^{tar-PLC}\}} 1_{\{S_{t+1} \leq S_{i+1}^{tar-PLC}\}} \\ & + \sum_{i=1}^{t-1} P(t-i) 1_{\{S_t > S_{i+1}^{tar-PLC}\}} 1_{\{S_{t+1} \leq S_{i+1}^{tar-PLC}\}} + 1_{\{S_{t+1}=0\}} 1_{\{NI_t^{PLC} < 0\}} NI_t^{PLC}. \end{aligned}$$

Since $S_t > S_t^{tar-PLC}$ and by setting $S_1^{tar-PLC} = S_1$, the inequality $S_j^{tar-PLC} \geq S_t > S_{j+1}^{tar-PLC}$ must hold for some $j \in \{1, \dots, t-1\}$. Then, when $S_{j+1}^{tar-PLC} \geq S_{t+1} > S_{j+2}^{tar-PLC}$, we have

$$h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t) = P(t-j) > 0.$$

When $S_j^{tar-PLC} \geq S_{t+1} > S_{j+1}^{tar-PLC}$, we have

$$h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t) = 0.$$

Thus, the current penalty function is a nonincreasing function of S_{t+1} .

c) When $t = D$ and $S_D > 0$, we have

$$\begin{aligned} h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t) = & \sum_{i=1}^D P(t-i) 1_{\{S_t > S_{i+1}^{tar-PLC}\}} 1_{\{S_{t+1} \leq S_{i+1}^{tar-PLC}\}} \\ & + 1_{\{S_{t+1}=0\}} 1_{\{NI_t^{PLC} < 0\}} NI_t^{PLC} + 1_{\{S_{t+1}>0\}} 1_{\{NI_t^{PLC} > 0\}} NI_t^{PLC}. \end{aligned}$$

When $t > D$ and $S_t > 0$, we have

$$h_{PLC}(S_{t+1}; S_t, NI_t^{PLC}, t) = \sum_{i=1}^D P(t-i) 1_{\{S_t > S_{i+1}^{tar-PLC}\}} 1_{\{S_{t+1} \leq S_{i+1}^{tar-PLC}\}}.$$

Since $S_t > 0 = S_{D+1}^{tar-PLC}$ and by setting $S_1^{tar-PLC} = S_1$, the inequality $S_j^{tar-PLC} \geq S_t > S_{j+1}^{tar-PLC}$ must hold for some $j \in \{1, \dots, D\}$. Applying the same argument as in part b), we conclude that the current penalty function is a nonincreasing function of S_{t+1} .